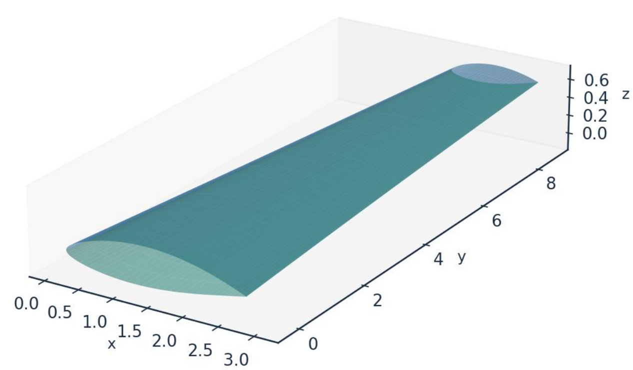
INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, M=0.42, Re_MAC ≈ 6.4×10^6):

Quantity	Predicted value
Upper-surface transition x_tr/c	0.554 (TS-natural)
Lower-surface transition x_tr/c	0.578 (TS-natural)
Mean laminar-flow extent	58.3 % of chord
Section lift coefficient C_l	0.517
Section profile drag C_d	43.8 counts
Viscous drag reduction vs fully-turbulent	50.4 %
Climb (Tu=0.9 %) transition x_tr/c (upper)	0.025 (bypass)

Table 1. Headline predictions, read from the generated solution CSVs.

Validated against eight independent, credible published datasets with one universal calibration set — four ERCOFTAC flat plates (case 020), the Schubauer-Skramstad plate, the NLF(1)-0416 aerofoil at 86 conditions, and two swept wings — the solver reproduces the transition-onset Reynolds number Re_θt and the measured transition location without per-case re-tuning of the physics. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, predicted here by an e^N integral carried per physical frequency, with the spatial growth rates read from a tabulated solution of the Orr-Sommerfeld problem rather than from an envelope correlation.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent. The bubble is closed explicitly here — the shear layer is carried across the dead-air region by the momentum integral and reattachment placed where it has amplified by the same critical factor used elsewhere — so a length, and not merely a separation point, is predicted.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f, momentum thickness θ, shape factor H, intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e, with a Karman-Tsien correction applied to C_p and the integrated loads. It returns a stagnation pressure coefficient of 1.048 at M = 0.42 against the exact isentropic 1.045, where the linearised Prandtl-Glauert scaling gives 1.102, and the two agree to better than half a per cent below M = 0.2.
- Laminar boundary layer: the momentum AND kinetic-energy integral equations are advanced together from the stagnation point along each surface, so the shape factor is a solved variable carrying its own history rather than a local function of the pressure gradient. The three closure functions - H*(H), Re_theta*Cf/2 and Re_theta*C_D - are properties of

the Falkner-Skan family, computed from it and not fitted; every similarity solution is an exact fixed point of the second equation, and the march returns H = 2.5914 on a flat plate against the Blasius 2.5913. Fluid properties are evaluated at Eckert's reference temperature so that the incompressible closures return the compressible skin friction.

- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ.
- Turbulent boundary layer: Head's entrainment method with the Ludwieg–Tillmann skin-friction law advances the turbulent BL to the trailing edge.
- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

Two elements of the formulation are not correlations. The amplification rate driving the e^N integral is tabulated in solver/amplification_db.npz from 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, continued past separation onto its reverse-flow branch: profiles are generated by continuation, the eigenvalue problem is solved by Chebyshev collocation, and the spatial growth rate is recovered from the temporal one through Gaster's transformation. The stability solver returns the Blasius neutral point at Re_theta = 200 against the accepted 200.5, a shape factor of 2.59129 and a wall shear parameter f''(0) = 0.46960. At run time the cost is a table lookup, so the sub-second solution is preserved. The second is the separation-bubble closure described above, whose length scales with the disturbance environment and therefore spans measured bubbles from about 40 momentum thicknesses at 2.1 % free-stream turbulence to about 180 at 0.03 %.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) Vortex-panel surface velocity (Kuethe & Chow)

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^{N} C_{t,ij}\gamma_j$$

(E03) Kutta condition (sharp trailing edge)

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) Pressure coefficient (inviscid)

$$C_p = 1 - \left(\frac{V}{U_\infty}\right)^2$$

(E04) Karman-Tsien compressibility correction

$$C_p = \frac{C_{p,0}}{\beta + \left(\frac{M_\infty^2}{1+\beta}\right)\frac{C_{p,0}}{2}}, \beta = \sqrt{1 - M_\infty^2}$$

4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6}\int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu}\frac{dU_e}{dx}, Re_\theta = \frac{U_e\theta}{\nu}$$

(E06b) Kinetic-energy integral (two-equation laminar march)

$$\theta\frac{dH^*}{dx} = 2C_D - H^*\frac{C_f}{2} - H^*(1-H)\frac{\theta}{U_e}\frac{dU_e}{dx}$$

(E06c) Laminar closure functions (from the Falkner-Skan family)

$$H^*(H) = \frac{\theta^*}{\theta}, l(H) = Re_\theta\frac{C_f}{2} = \theta_\eta f''(0), d(H) = Re_\theta C_D = \theta_\eta\int (f'')^2 d\eta$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2+H)\frac{\theta}{U_e}\frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the flow-history-averaged Tu; natural/TS onset integrates one amplification factor per physical frequency using the tabulated Orr-Sommerfeld growth rates and triggers on their envelope at N_crit; separation-induced onset closes a laminar bubble across the dead-air region; and cross-flow onset uses the C1 criterion. The kernel takes the minimum effective onset across all four:

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp\left[F(\lambda_\theta) - \frac{F(\lambda_\theta)Tu}{6.91}\right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \le 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (e^N, envelope over frequency)

$$N(x) = \max_\omega \int_{x_0(\omega)}^x \frac{\sigma(H, Re_\theta, \omega\theta/U_e)}{\theta} dx' \ge N_{crit}$$

(E10b) Orr-Sommerfeld operator (tabulated amplification rates)

$$[(U-c)(D^2-\alpha^2)-U'']\hat{v}=\frac{1}{i\alpha Re_\theta}(D^2-\alpha^2)^2\hat{v}$$

(E10c) Gaster transformation (temporal to spatial growth rate)

$$\sigma=-\alpha_i\theta=\frac{\omega_i}{c_g},c_g=\frac{\partial\omega_r}{\partial\alpha_r}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2}=k_{cf}Re_\theta\sin\Lambda\cos\Lambda\ \geq\ C_1$$

(E11b) Cross-flow amplification integral (the closure actually used)

$$N_{cf}=\int_{x_{c1}}^{x}\frac{\sigma(H_{rev})}{\theta}dx'\ \geq\ N_{crit}$$

(E12) Separation bubble: dead-air march (momentum and kinetic energy, no wall shear)

$$\frac{d\theta}{dx}=-(2+H)\frac{\theta}{U_e}\frac{dU_e}{dx},\theta\frac{dH^*}{dx}=2C_D-H^*(1-H)\frac{\theta}{U_e}\frac{dU_e}{dx},C_f=0$$

(E12b) Separation bubble: reattachment condition

$$N_{bub}=\int_{x_s}^{x_r}\frac{\sigma(H_{rev},Re_\theta)}{\theta}dx'=N_{crit},\sigma(H_{rev})\approx 0.0435$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^{*}=\min\left(a_{TS}Re_{\theta t}^{TS},a_{BP}Re_{\theta t}^{BP},a_{SEP}Re_{\theta t}^{SEP},a_{CF}Re_{\theta t}^{CF}\right)$$

(E14) Transition trigger

$$Re_\theta(x_t)\geq Re_{\theta t}^{*}$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x)=1-\exp[-0.412\xi^2],\xi=\frac{x-x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr}\sim\frac{\nu}{U_e}C_{len}Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi=(1-\gamma)\phi_{lam}+\gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx}=C_E-\frac{\theta H_1}{U_e}\frac{dU_e}{dx},C_E=0.0306(H_1-3)^{-0.6169}$$

(E19) Ludwieg-Tillmann skin-friction law

$$C_f=0.246\times 10^{-0.678H}Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2\frac{\theta_{TE}}{c}\left(\frac{U_{e,TE}}{U_\infty}\right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r\frac{\gamma-1}{2}M_e^2\left[1-\left(\frac{u}{U_e}\right)^2\right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e\left(1 + r\frac{\gamma-1}{2}M_e^2\right), r \approx Pr^{1/3}$$

(E22b) Eckert reference temperature (compressible closures)

$$\frac{T_{ref}}{T_e} = 1 + 0.032M_e^2 + 0.58\left(\frac{T_w}{T_e}-1\right), \frac{\nu_{ref}}{\nu_e} = \left(\frac{T_{ref}}{T_e}\right)^{1+\omega}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3}c_{root}\frac{1+\lambda+\lambda^2}{1+\lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e^N codes	RANS γ–Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (tabulated e^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble closure)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. Every one of the four branches is the selected mechanism somewhere in this study — natural/TS on the cruise wing, the Schubauer-Skramstad plate and the NLF(1)-0416 upper surface; bypass on the climb case and the ERCOFTAC plates; separation on T3C4 and the NLF(1)-0416 lower surface; cross-flow on both swept wings — and each is checked against measurement in Section 11.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mach	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (FL100, M0.3	3000.0	0.3	98.57	0.9093	1.694e-05	268.66	0.9	3.5	10700000.0	1.863e-05	4417.8

0)												

Table 4. Flight / flow conditions (input).

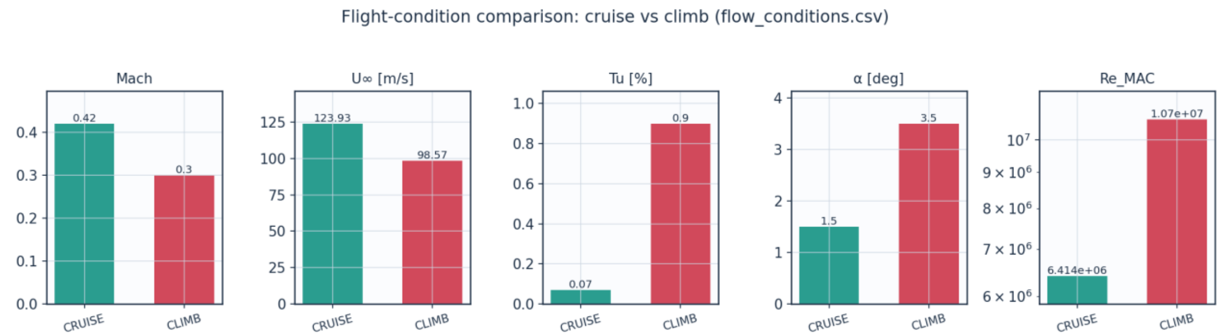


Fig. 8a. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H*, l and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6
Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)

Wall y+ target	~1 (reconstruction grid)

Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	unity; validated on Schubauer-Skramstad and NLF(1)-0416 upper surface
A_BP	1.0	bypass onset weight	unity; validated on ERCOFTAC T3A/T3A-/T3B
A_SEP	1.0	separation-induced weight	unity; validated on T3C4 and NLF(1)-0416 lower surface
A_CF	1.0	crossflow weight	unity; Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack N=-8.43-2.4 ln(Tu), clamped to 0.0008<=Tu<=0.0298, anchored -0.50 (FITTED jointly on S&S and 86 aerofoil pts); 8.18 for Tu<0.08%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
Tu_TS_max	0.1	upper Tu for the attached-flow TS branch	complement of the bypass gate
C_len	6.5	Narasimha transition-length scale	Dhawan-Narasimha (1958)
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al.; facility-dependent: 150 fits Dagenhart (14.7%), 200 fits Boltz (18.4%)
cf_amp	True	crossflow closed by an amplification integral, not a local threshold	rate computed (0.0435); threshold is the same N_crit; adds no constant
CF_ratio	0.47	theta2/theta surrogate for crossflow	FITTED on Dagenhart & Saric; independent check in Sec. IV.C
sigma_sep	from table	shear-layer amplification rate in a bubble	computed on the reverse-flow Falkner-Skan branch; 0.0435, not fitted
two_eq	True	two-equation laminar march (momentum + kinetic energy)	closures from the Falkner-Skan family; no fitted constant
N_anchor	0.5	offset on Mack's N_crit	unit conversion to the present amplification rates; set jointly on S&S and the 86 NLF(1)-0416 conditions
tu_hist	0.6	weight on the flow-history average of Tu, Tu_eff = Tu_avg^w Tu_local^(1-w)	FITTED on the three ERCOFTAC plates; no effect where Tu does not decay
lam_sep	-0.09	Thwaites parameter at laminar separation	Thwaites (1949) published value; not fitted
sep_floor	120.0	min separation-induced onset Re_theta (ablation path only)	superseded by the bubble closure; reachable only with bubble=False
n_freq	40	frequencies carried by the e^N integration	discretisation parameter; onset moves <0.2% over 24-64

Table 7. Calibration constants and their sources (input).

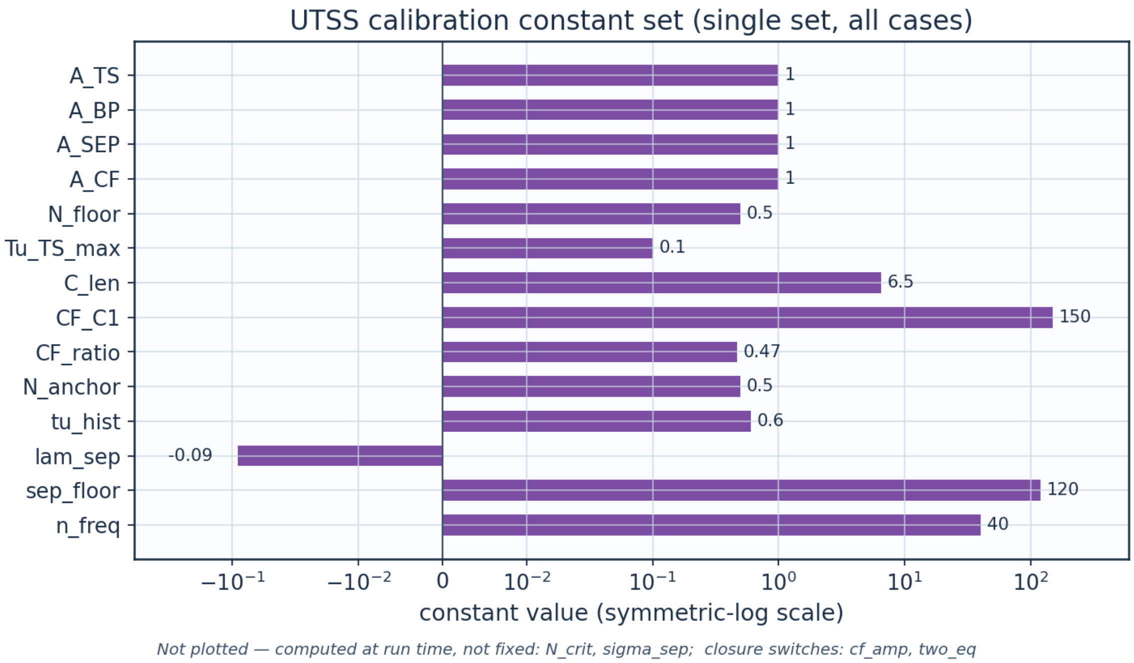


Fig. 8b. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

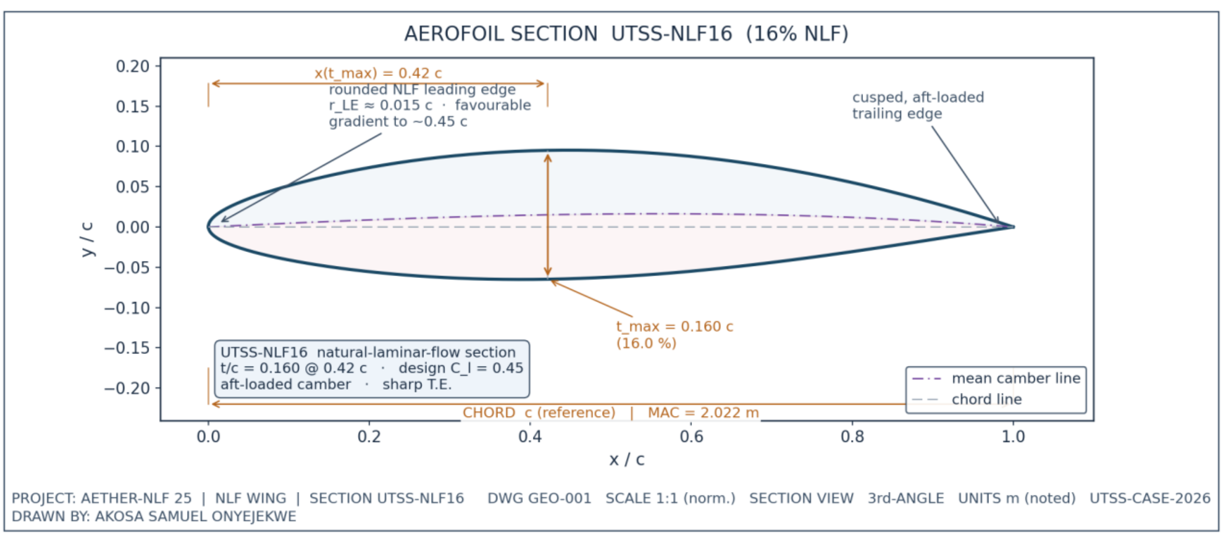


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

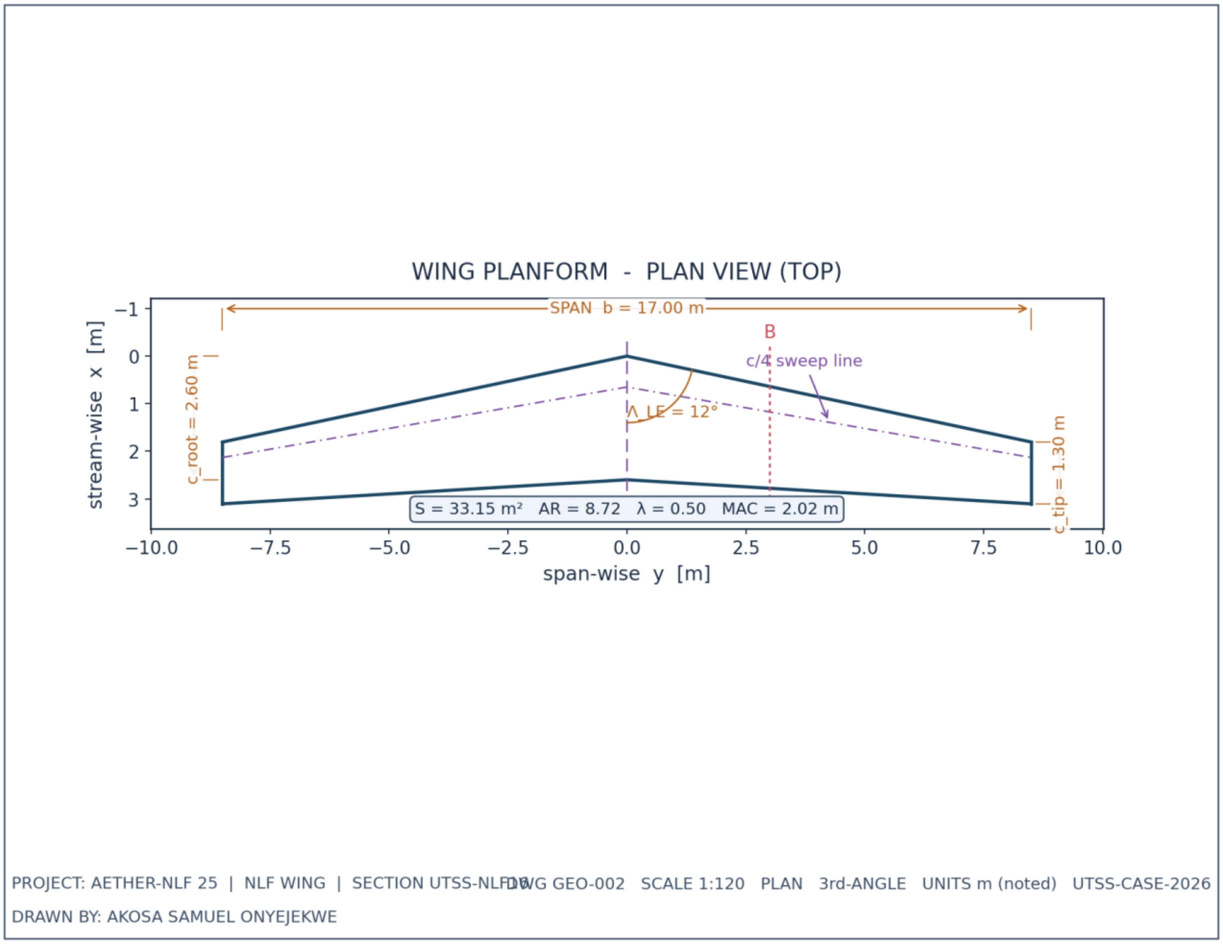


Fig. 2. Wing planform — plan (top) view, dimensioned.

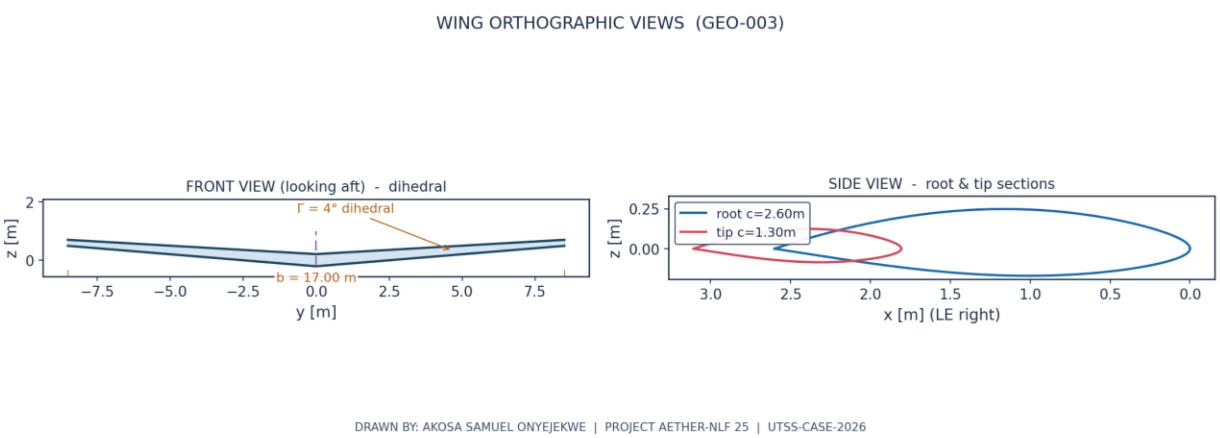


Fig. 3. Front and side orthographic views.

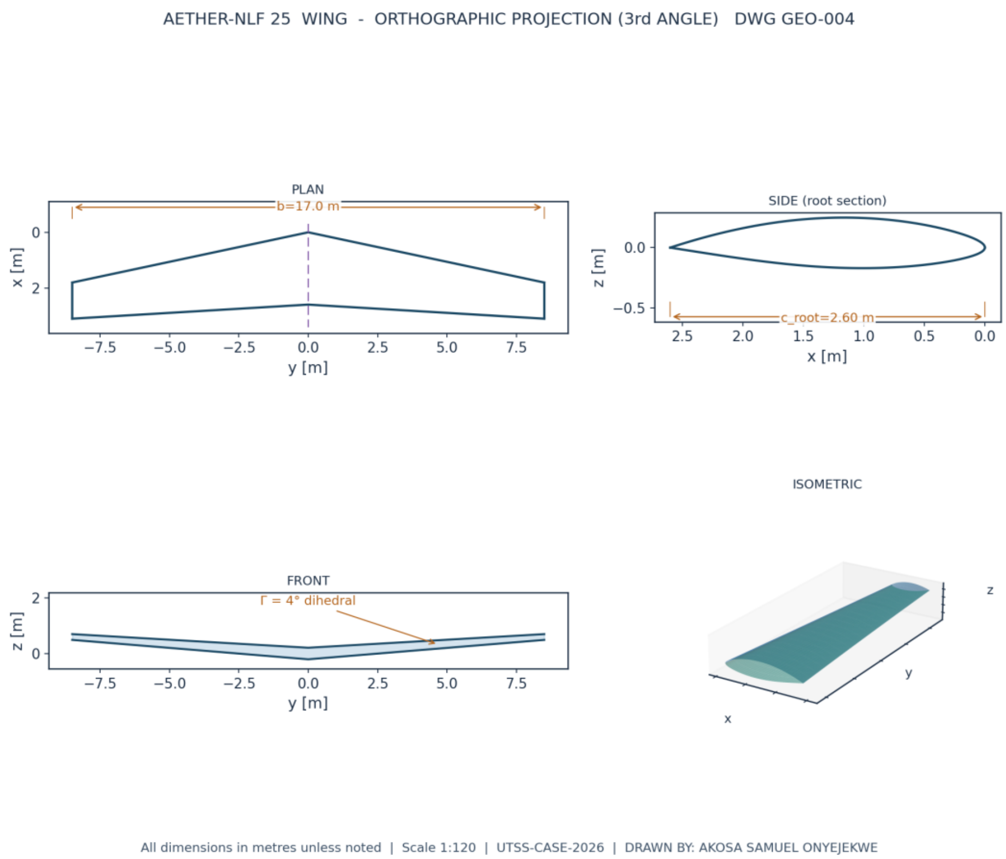


Fig. 4. Full third-angle orthographic projection sheet.

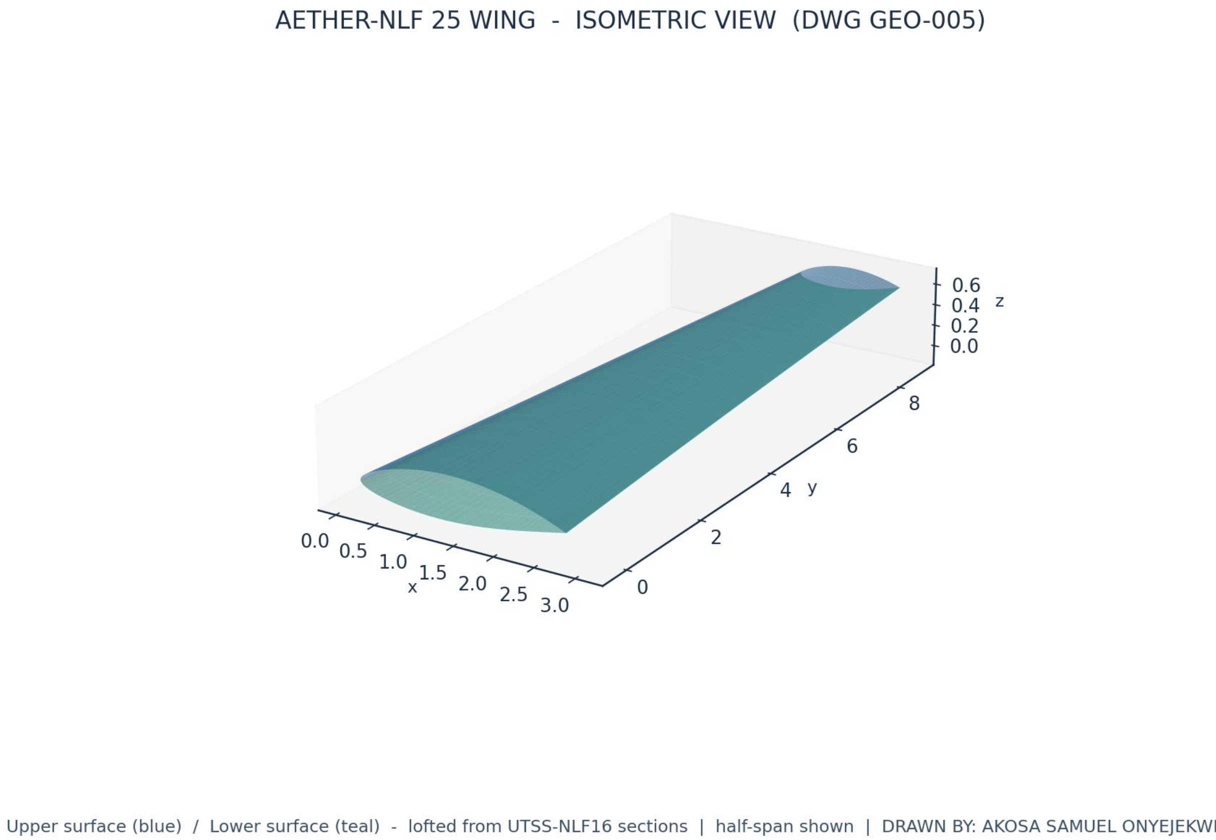


Fig. 5. Isometric view of the wing.

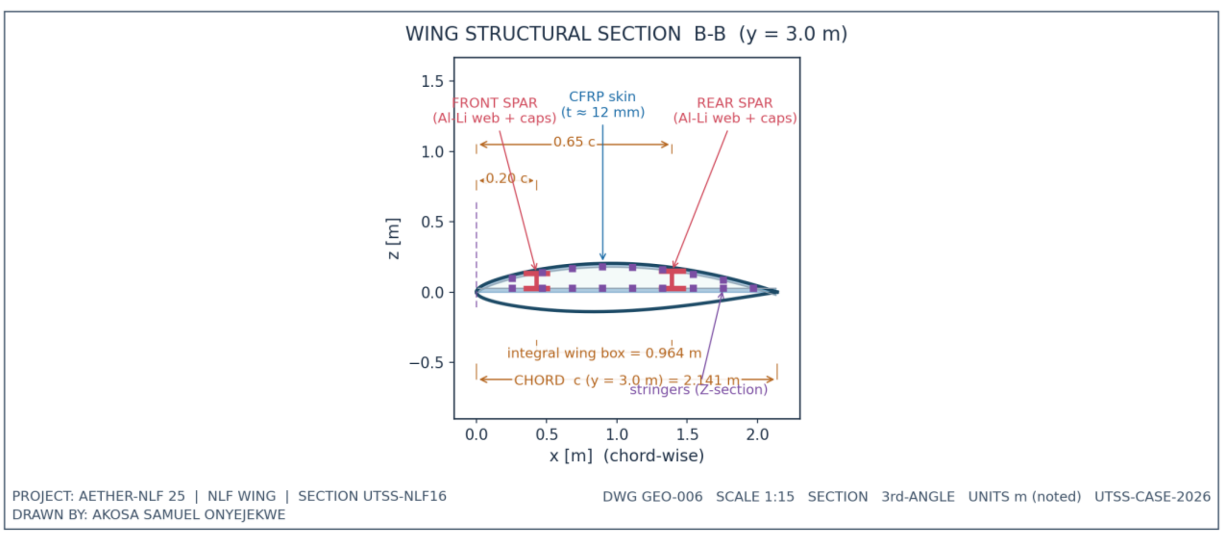


Fig. 6. Structural sectional view B–B.

7.1 Geometry data (from CSV)

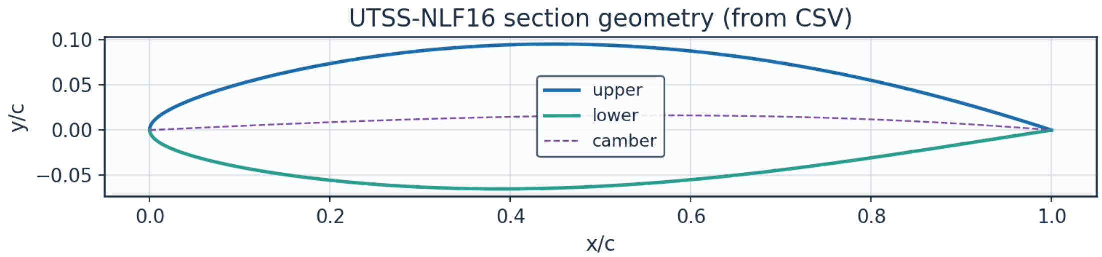


Fig. 7. Section geometry plotted from airfoil_UTSS-NLF16.csv.

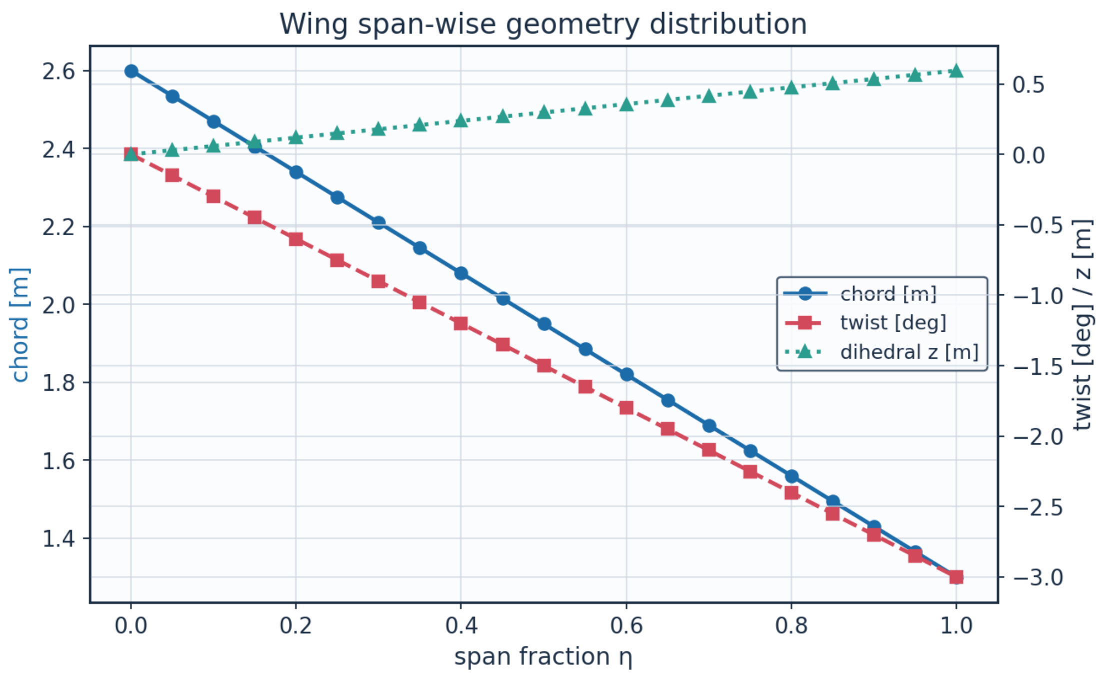


Fig. 8. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m	Re_local
0.0	0.0	2.6	0.0	2.6	-0.0	0.0	8246203.012995781
0.05	0.425	2.535	0.0903365387097594	2.62533653870976	-0.15	0.0297188950759919	8040047.9376708865
0.1	0.8500000000000001	2.47	0.1806730774195188	2.650673077419519	-0.3	0.0594377901519838	7833892.862345993
0.15	1.275	2.405	0.2710096161292782	2.6760096161292783	-0.45	0.0891566852279757	7627737.787021098
0.2	1.7000000000000002	2.34	0.3613461548390376	2.7013461548390376	-0.6000000000000001	0.1188755803039677	7421582.711696202
0.25	2.125	2.275	0.451682693548797	2.726682693548797	-0.75	0.1485944753799596	7215427.636371308
0.3	2.5500000000000003	2.21	0.5420192322585565	2.7520192322585566	-0.9000000000000001	0.1783133704559515	7009272.561046414

0.35	2.975	2.145	0.6323557709683159	2.777355770968316	-1.05	0.2080322655319434	6803117.485721519
0.4	3.4000000000000004	2.08	0.7226923096780753	2.802692309678076	-1.2000000000000002	0.2377511606079354	6596962.410396625
0.45	3.825	2.015	0.8130288483878346	2.828028848387835	-1.35	0.2674700556839273	6390807.33507173
0.5	4.25	1.95	0.903365387097594	2.853365387097594	-1.5	0.2971889507599192	6184652.2597468365
0.55	4.675000000000001	1.885	0.9937019258073536	2.878701925807354	-1.65	0.3269078458359112	5978497.184421942
0.6000000000000001	5.1000000000000005	1.82	1.084038464517113	2.9040384645171127	-1.8000000000000005	0.3566267409119031	5772342.109097046
0.65	5.525	1.755	1.1743750032268725	2.9293750032268724	-1.95	0.386345635987895	5566187.033772152
0.7000000000000001	5.95	1.69	1.2647115419366315	2.954711541936632	-2.1	0.4160645310638869	5360031.958447257
0.75	6.375	1.625	1.3550480806463912	2.980048080646391	-2.25	0.4457834261398788	5153876.883122363
0.8	6.800000000000001	1.56	1.4453846193561506	3.00538461935615	-2.4000000000000004	0.4755023212158709	4947721.807797468
0.8500000000000001	7.225	1.495	1.53572115806591	3.03072115806591	-2.5500000000000003	0.5052212162918628	4741566.732472573
0.9	7.65	1.43	1.6260576967756692	3.056057696775669	-2.7	0.5349401113678547	4535411.65714768
0.95	8.075000000000001	1.365	1.716394235485429	3.081394235485429	-2.85	0.5646590064438467	4329256.581822785
1.0	8.5	1.3	1.806730774195188	3.106730774195188	-3.0	0.5943779015198385	4123101.50649789

Table 8. Wing planform stations (wing_planform.csv).

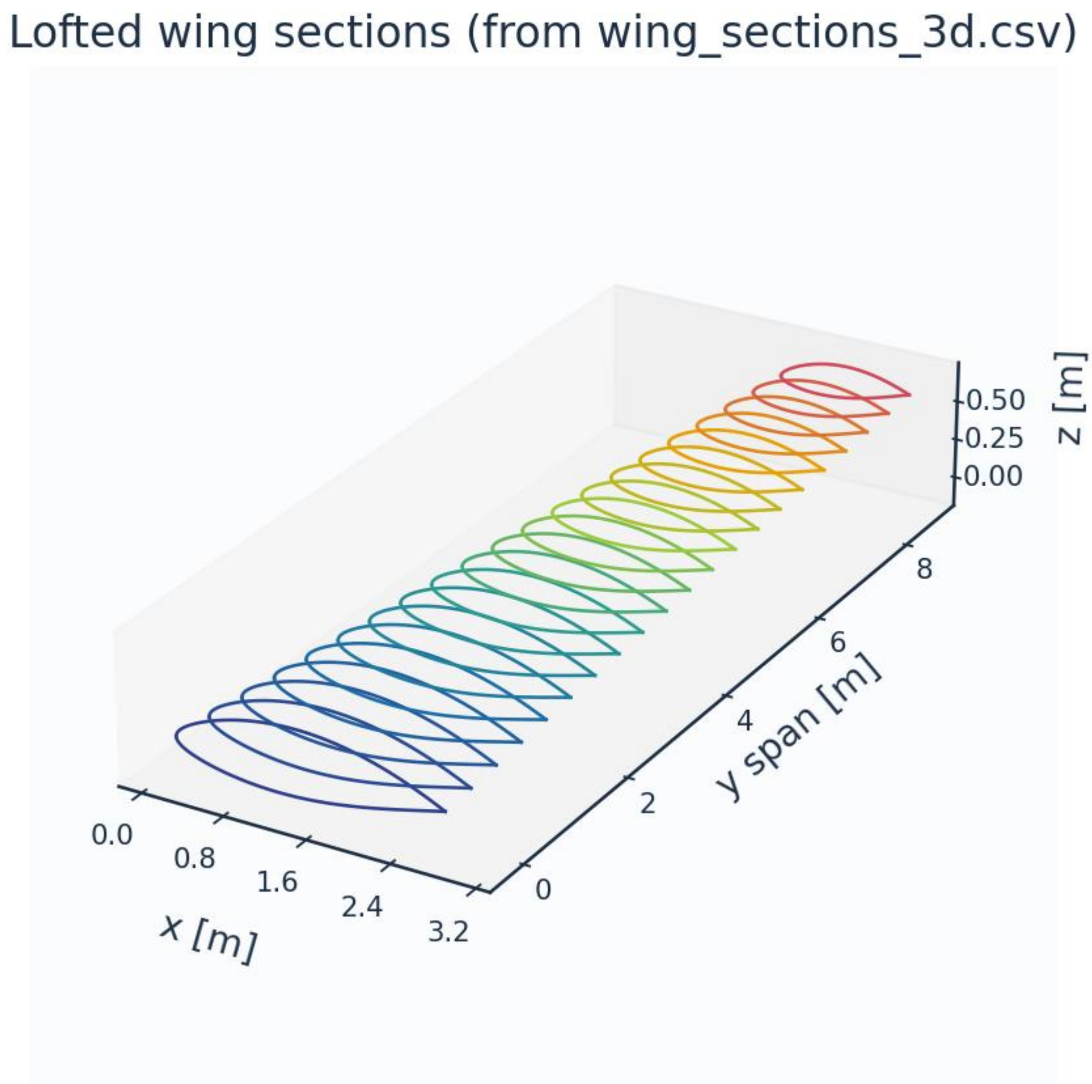


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^+ \approx 1$) supports boundary-layer profile recovery. A panel-count sweep bounds the discretisation sensitivity rather than demonstrating asymptotic convergence: from 120 to 480 surface panels the section drag spans 2.1 counts, 4.7 % of its mean, and above 180 panels it stays within about ±0.5 count without tightening further. The residual wander is not a truncation error that refinement removes — it is set by which panel the transition point lands on, so it scales with the panel spacing at transition and is of the same order as the ±0.025c bracket the aerofoil measurements themselves carry. The 260-panel grid is used throughout.

metric	value	unit

Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.5146538348994995	0.0042584577133717	0.538986194980472	
180	0.5161840825332155	0.0043773257315635	0.5433865690013859	2.791339639666557
260	0.5170864702757358	0.0043837763667638	0.5541908676652394	0.1473647518120291
360	0.5176435398599823	0.0044661036655082	0.5477820283105952	1.8779995113031636
480	0.5180064091537756	0.0044403358876658	0.5554577472375675	-0.5769632720663997

Table 10. Panel-count sensitivity study.

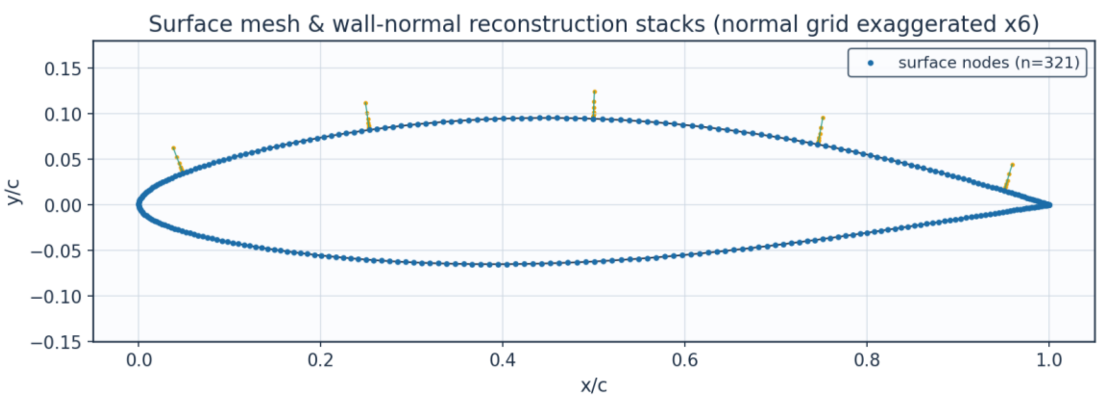


Fig. 9. Surface mesh and wall-normal stacks.

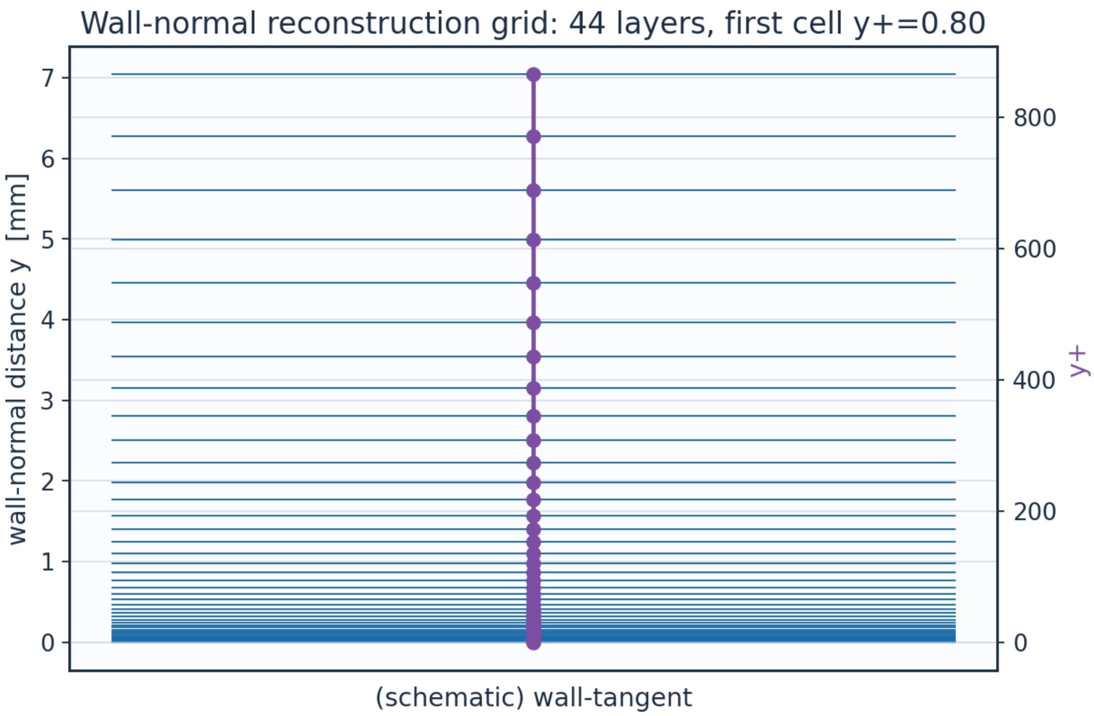


Fig. 10. Wall-normal reconstruction grid / y⁺.

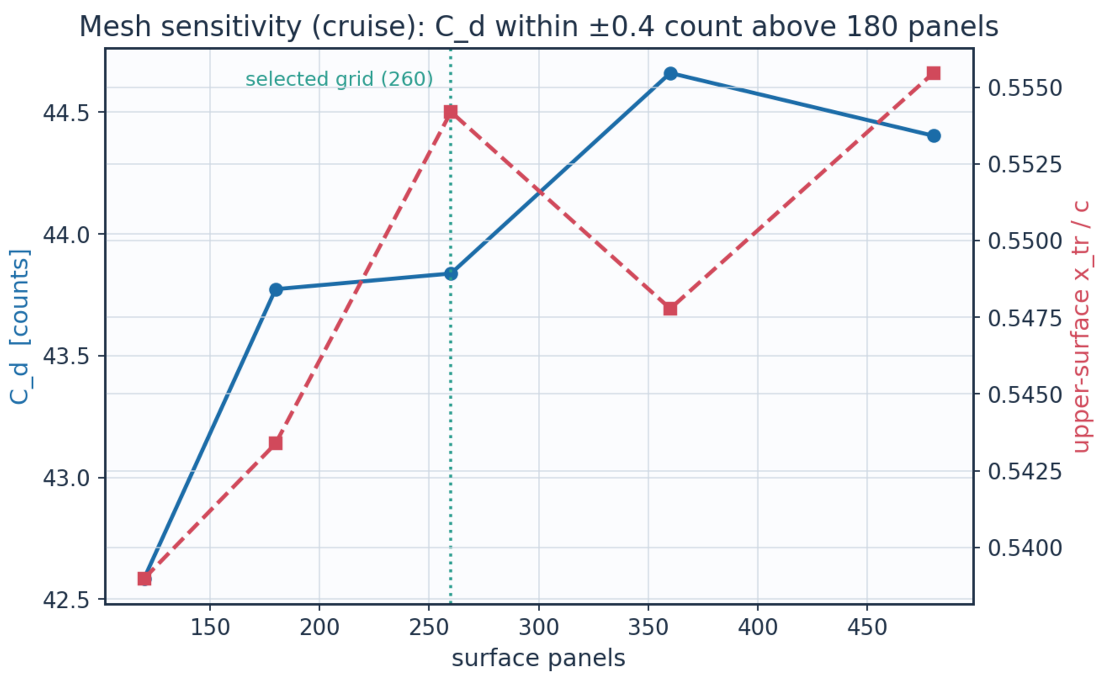


Fig. 11. Panel-count sensitivity of C_d and transition location.

layer	y_m	y_plus	cell_dy_m	growth_ratio
0.0	0.0	0.0	6.5127391337287455e-06	1.0
2.0	1.380700696350494e-	1.696	7.731923899562769e-	1.12

	05		06	
4.0	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6.0	5.2852109259255385e-05	6.4921512345600005	1.2166331946008715e-05	1.12
8.0	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10.0	0.0001142903336347	14.038988055628067	1.9143958857231197e-05	1.1199999999999997
12.0	0.0001571728014749	19.30650661697985	2.401418199051081e-05	1.12
14.0	0.0002109645691337	25.91408190033953	3.012338988889677e-05	1.12
16.0	0.0002784409624848	34.20262433578592	3.778678027663213e-05	1.120000000000001
18.0	0.0003630833503045	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20.0	0.0004692587615855	57.64195395516631	5.945823031734682e-05	1.12
22.0	0.0006024451974963	74.00206704136062	7.458440411007986e-05	1.12
24.0	0.0007695142627029	94.52419289668278	9.355867651568416e-05	1.12
26.0	0.000979085698098	120.26714756959888	0.0001173600038212	1.1200000000000006
28.0	0.0012419721066577	152.55910991130486	0.0001472163887934	1.1200000000000008
30.0	0.0015717368175549	193.0661474727408	0.0001846682381024	1.1199999999999997
32.0	0.0019853936709044	243.87817538980616	0.0002316478378757	1.1200000000000003
34.0	0.002504284827746	307.6167832089729	0.0002905790478312	1.119999999999999
36.0	0.0031551818948881	387.57049285733575	0.0003645023575995	1.12
38.0	0.0039716671759111	487.864426240242	0.0004572317573729	1.12
40.0	0.0049958663124264	613.6731362757596	0.0005735515164485	1.1200000000000014
42.0	0.0062806217092712	771.4875821443129	0.000719463022233	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism	laminar_run_pct	Cf_te
CRUISE	upper	0.579	0.554	3712000.0	1410.4	TS-natural	55.4	0.00021
CRUISE	lower	0.587	0.578	3766000.0	1270.7	TS-natural	57.8	0.00025
CLIMB	upper	0.047	0.025	506900.0	483.8	bypass	2.5	0.00014
CLIMB	lower	0.159	0.157	1706000.0	610.7	bypass	15.7	0.00019

Table 12. Transition prediction summary (transition_summary.csv).

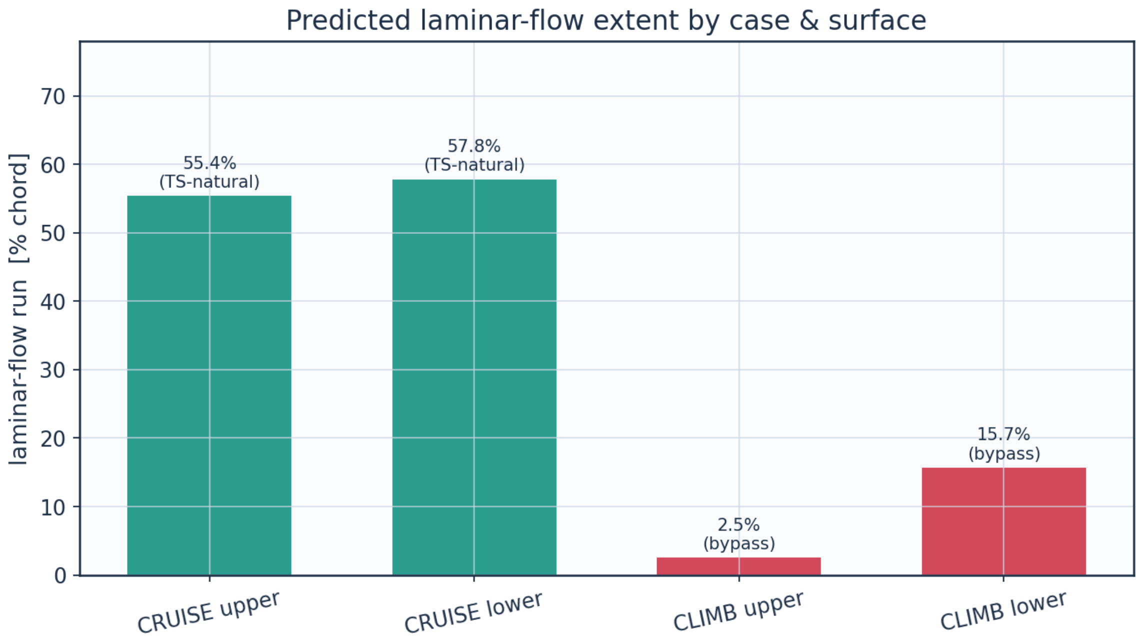


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00438	-
Section Cd (counts)	43.8	counts
Section L/D	118.0	-
Wing lift (3-D estimate, C_L = 0.90 C_l)	43114.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Table 13. Integrated forces.

configuration	Cd_counts	mean_laminar_pct
NLF (UTSS predicted transition)	43.8	58.3
Fully turbulent (LE trip)	88.4	0.0
Viscous drag reduction	50.4	

Table 14. NLF vs fully-turbulent drag.

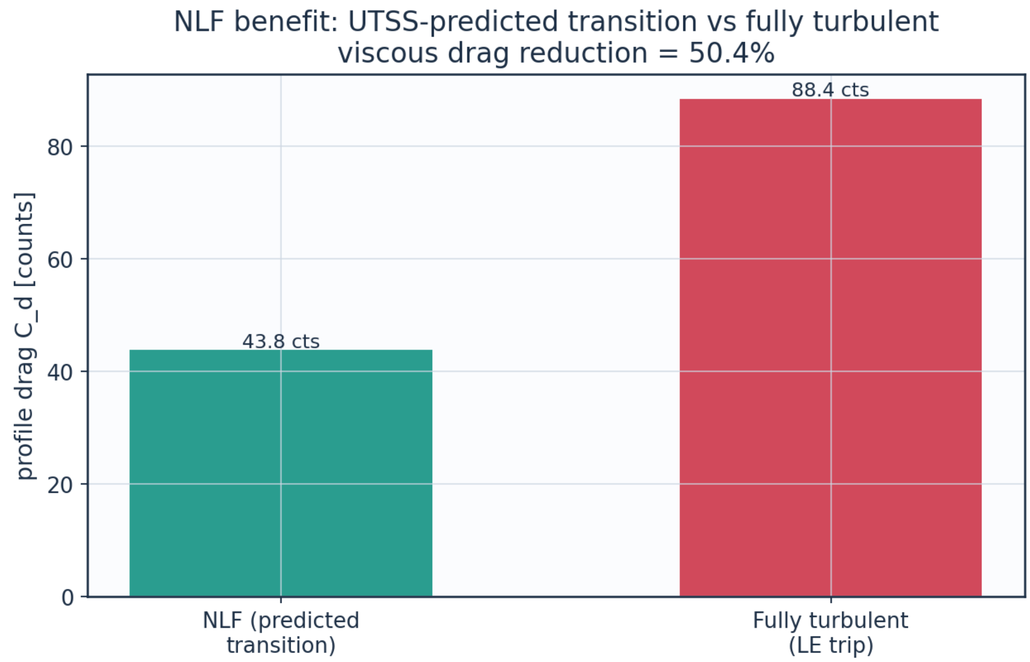


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.3 Surface distributions — cruise

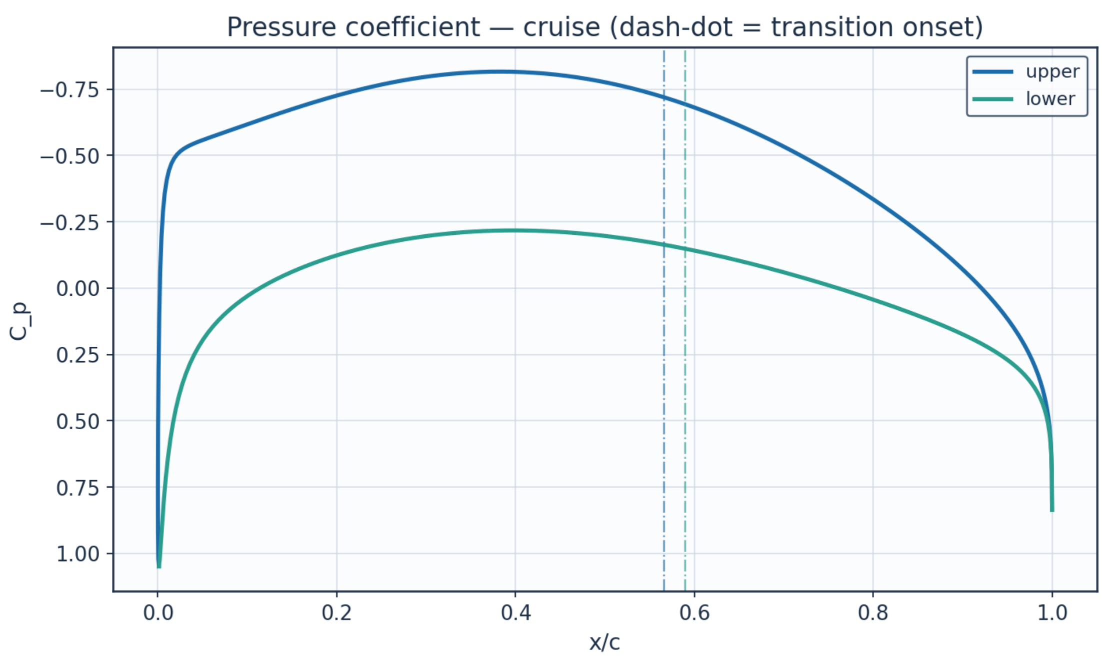


Fig. 13. Pressure coefficient C_p — cruise.

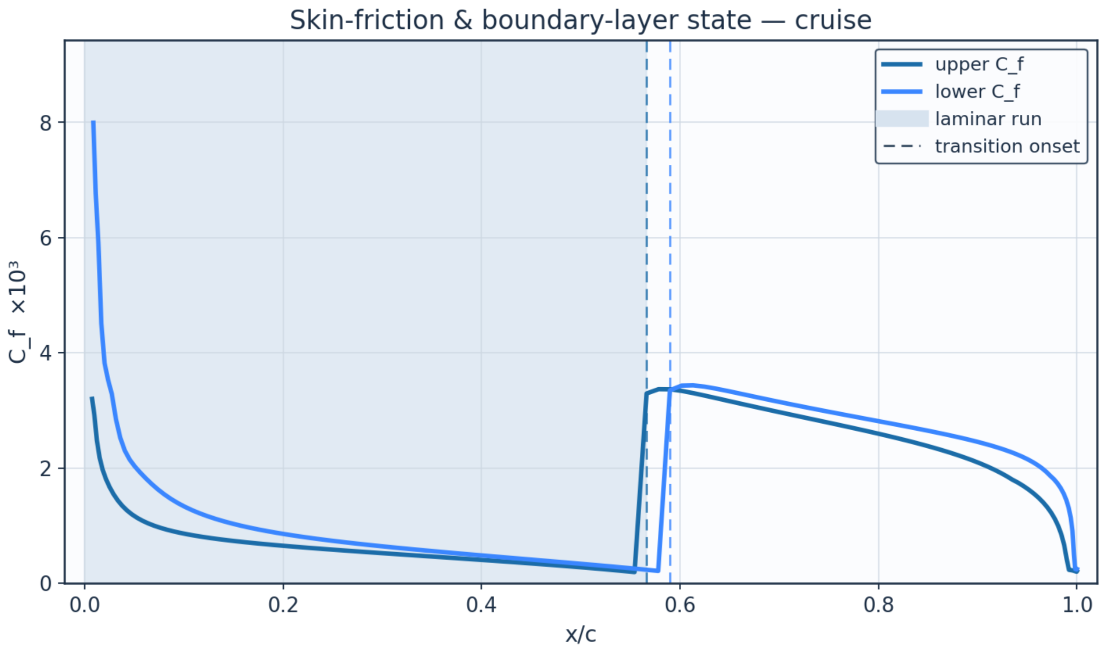


Fig. 14. Skin-friction C_f and laminar run — cruise.

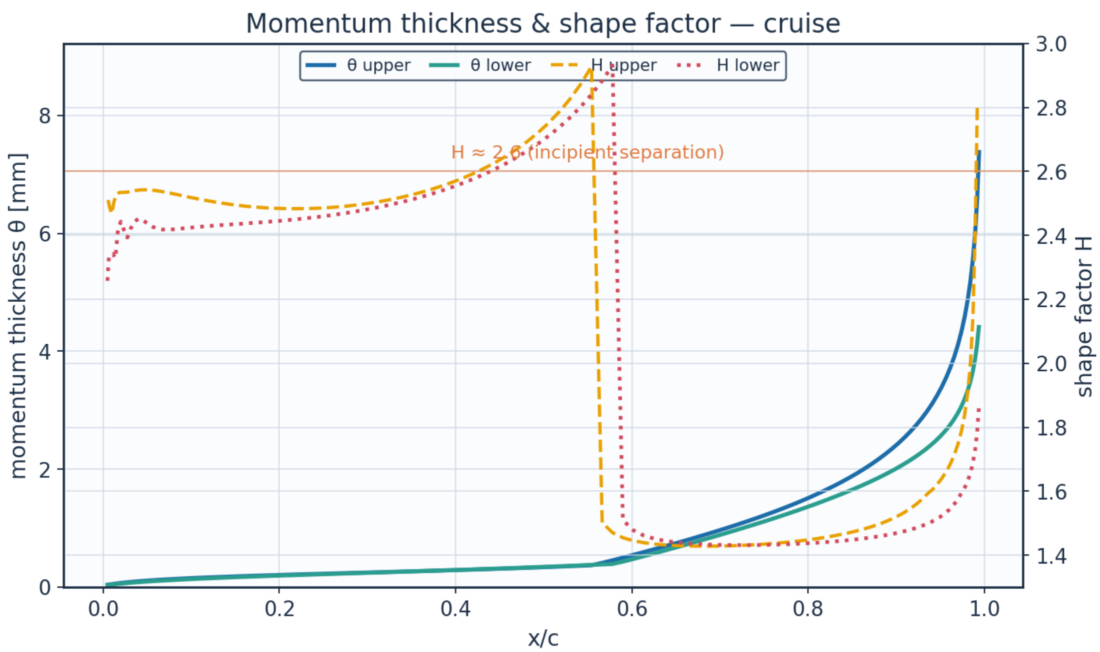


Fig. 15. Momentum thickness ϑ and shape factor H — cruise.

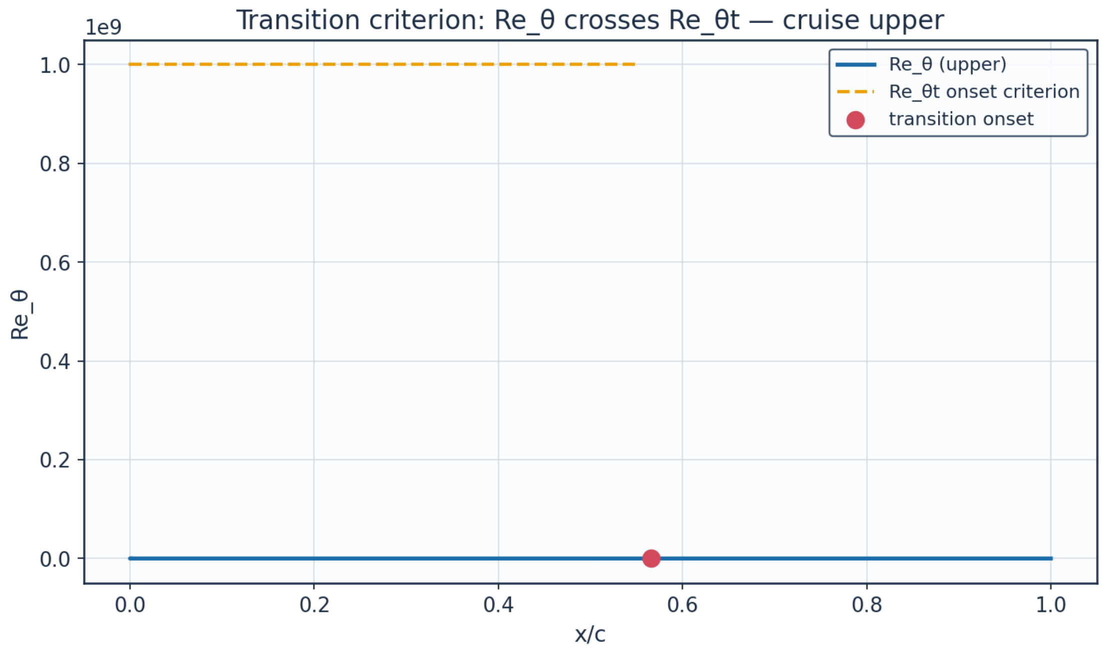


Fig. 16. Transition criterion Re_ϑ vs Re_ϑt — cruise.

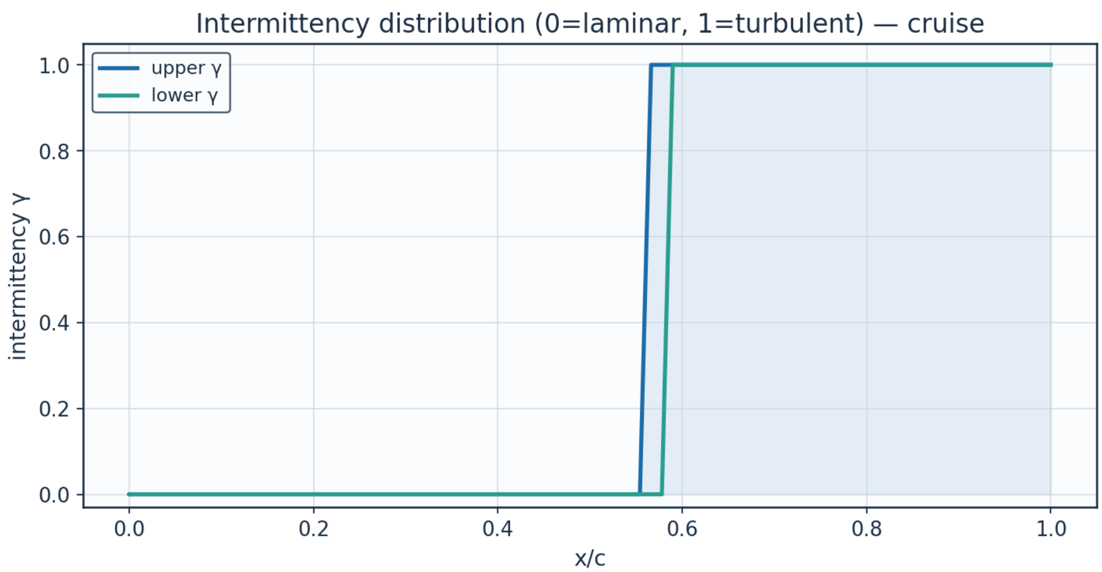


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011523200325028	0.0	0.0	1.0484500005807595	0.6743779759375519	0.0054416302825443	1e-05	2.6099999414507185	2491.908085201937	0.0001725874393307	1000000000.0	0.0	laminar
0.000242727	0.014097598	44712.17719	0.531909959	88.07774049	0.710709004	0.026313920	2.131948169	0.013827901	58.12010330	10000000	0.0	lamina

1046013	6799438	636864	2569656	458455	4378861	3079384	0088806	7318326	748996	00.0		r
0.0039984634929024	0.0303574508340593	96282.19328257376	-0.1906595115746616	134.13696378207842	1.082365958215552	0.037378612170277	2.462407997730696	0.0042799774898428	122.44703974006676	1000000000.0	0.0	laminar
0.0123828259304446	0.052871121292092	167686.92296127492	-0.4431856098056879	146.32725910183538	1.1807307959357134	0.0562800876810639	2.505801090933648	0.0024818494066748	199.367350794746	1000000000.0	0.0	laminar
0.0253172774881581	0.0831882375254351	263841.19035694306	-0.5161572579089404	149.61652550936128	1.2072722494368673	0.0791442822357182	2.534326561988456	0.0016657296805269	285.952514374348	1000000000.0	0.0	laminar
0.0426830314241258	0.1217109964051874	386020.6097196828	-0.5481704273505028	151.03025162861496	1.2186797614497848	0.1023649543614522	2.541964874436309	0.0012640951244638	372.9402935748207	1000000000.0	0.0	laminar
0.0643227586523211	0.1683869080802935	534058.7033695233	-0.5751048472060674	152.20640645138116	1.2281702844634217	0.1246539560144825	2.5379251173937782	0.0010366343638922	457.26533154868616	1000000000.0	0.0	laminar
0.0900422957300214	0.2229351474645065	707064.8018171047	-0.6054032741270788	153.51532585711524	1.2387320995430888	0.1455636235289966	2.523781735915112	0.0008978756491207	538.0110117597337	1000000000.0	0.0	laminar
0.1196122456699886	0.2849313998941607	903693.1416555132	-0.6399761678473315	154.99111243577192	1.250640384249193	0.165128997834491	2.507577796128069	0.0008020183726366	615.4791217278166	1000000000.0	0.0	laminar
0.1527694018013956	0.353848290318842	1122271.0914502463	-0.6773382556320362	156.56525350268146	1.2633423021710868	0.1835810495681587	2.493799043571883	0.0007282351136715	690.3424030450623	1000000000.0	0.0	laminar
0.1892179872734863	0.4290798210019653	1360876.6587546526	-0.7150952205439175	158.13484523358554	1.2760075109988878	0.2012557344703509	2.4852163093874604	0.00066606040661	763.4345155900454	1000000000.0	0.0	laminar
0.2286307839316745	0.5099588956971477	1617394.0700776232	-0.7504707657473884	159.5866938991024	1.2877226380431313	0.2185549060746073	2.4833880267323125	0.000609960237384	835.6881399908314	1000000000.0	0.0	laminar
0.2706503108546585	0.59577083610005	1889556.6398858647	-0.7805852261678627	160.80872570370784	1.2975833474841956	0.2359240306060316	2.4896858991225392	0.0005566024948537	908.107089093837	1000000000.0	0.0	laminar
0.3148902862023041	0.6857636986922375	2174979.491676561	-0.80268469982041	161.69756919660222	1.3047555236820498	0.2538395197093098	2.5059865978013764	0.0005038714709239	981.752635784148	1000000000.0	0.0	laminar

			01									
0.360937646562712	0.7791556260346346	2471182.87346131445	-0.81436445232152126	162.16465934647778	1.308524525628929	0.2728102135724747	2.5344425388576663	0.0004502370758732	1057.7657970797738	1000000000.0	0.0	laminar
0.40835539017306884	0.875139546452957	2775607.063366206	-0.8137747395394551	162.1411197751585	1.3083345822311616	0.2933831276597855	2.578894097437993	0.0003943592017089	1137.3899656195374	1000000000.0	0.0	laminar
0.4566864473674744	0.9728858705600144	3085620.9219665444	-0.7997780647072827	161.58104307776875	1.3038152615744834	0.3161679242083832	2.64685310153594	0.0003346539138878	1222.050272458086	1000000000.0	0.0	laminar
0.5054586679985256	1.071544176368635	3398527.2368188924	-0.7720279102211003	160.4627537071297	1.2947916612775476	0.3418852992987908	2.7510967434132056	0.0002690140312561	1313.5072677340183	1000000000.0	0.0	laminar
0.5541908676652394	1.1702450637414947	3711568.604064828	-0.7309532082938842	158.7878833163172	1.2812769473290213	0.3714716877699943	2.927415498985632	0.0001929319314254	1410.3601916198566	1000000000.0	0.0	transitional
0.6023997176209082	1.268103309912791	4021937.436535685	-0.6776530070765481	156.57842511702646	1.2634485853802768	0.5502736013011976	1.4446862826757936	0.0033350218775895	2063.9454267791657		1.0	turbulent
0.649607126440503	1.3642231699528122	4326792.774793668	-0.6137239957904199	153.8722197384024	1.2416119156423129	0.7439155351456935	1.430109360007382	0.0031597941587687	2748.134247777037		1.0	turbulent
0.6953476717367839	1.4577062094846656	4623285.129505811	-0.5410537276879176	150.71747104689848	1.216155900431201	0.9453952040935488	1.4283904626434842	0.0029856753998762	3429.5567884358743		1.0	turbulent
0.7391756157699833	1.547661543814744	4908588.956039555	-0.461612518264526	147.16687039720225	1.1875057121006174	1.1589757285787652	1.4326637308514742	0.0028241353578541	4116.855580484189		1.0	turbulent
0.7806710851424759	1.633217905018701	5179940.926632257	-0.3772690650906994	143.27174669823822	1.1560755292790752	1.3896934014239697	1.441722299531118	0.0026692246788159	4820.233993732802		1.0	turbulent
0.8194451029680374	1.7135366561310357	5434681.206410178	-0.2896428230044323	139.0764000663416	1.1222228144922963	1.643503375277745	1.4557642256788652	0.0025143867572352	5551.1617996127225		1.0	turbulent
0.855143310574868	1.787824762773985	5670294.59442127	-0.1999936928488182	134.61243710955054	1.0862026049472566	1.9281290810651412	1.4758939932322108	0.0023529435271138	6324.090035258371		1.0	turbulent

0.8874483761131826	1.855346811455994	5884448.641069436	-0.10913820000032198	129.89245152018665	1.048116520536585	2.2545457462536804	1.504408045228816	0.0021769136508173	7159.296266550006		1.0	turbulent
0.916081229913384	1.9154353854551616	6075026.556438111	-0.0173695225920768	124.9015687555679	1.0078445369344795	2.6397456697507606	1.5458505779154907	0.0019749126875895	8087.936706777848		1.0	turbulent
0.9408013677638524	1.9674994072259613	6240153.669205437	0.0756600673579482	119.58356245557466	0.9649329574384664	3.112508952790904	1.6045025491377838	0.0017428686336219	9162.34067768806		1.0	turbulent
0.9614065112832176	2.0110303614800173	6378217.163870198	0.171181141306249	113.813455180805	0.9183733253029952	3.7256478813764926	1.685679107690956	0.0014812648985748	10475.959435836388		1.0	turbulent

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011523200325028	0.0	0.0	1.0484500005807595	0.6743779759375519	0.0054416302825443	1e-05	2.6099999692118865	2491.9079902448643	0.0001725874393307	1000000000.0	0.0	laminar
0.0067207749723691	0.0175589809676828	55690.354523400645	0.7967880591987423	61.8699583142874	0.4992355188864579	0.0406309338036424	2.337293310876968	0.0096899002459978	63.69016470587896	1000000000.0	0.0	laminar
0.0168974815052617	0.0419271025953991	132976.692210909	0.5091254824071559	89.95065248466179	0.7258217378980435	0.054733393189349	2.417731536216506	0.0045056870210784	123.35422644645288	1000000000.0	0.0	laminar
0.031587190340849	0.0742427608527724	235469.5685527914	0.3201161484737103	104.06544700919112	0.8397155720046343	0.0748295138803695	2.4226591611167576	0.0028504869710184	193.72467330060545	1000000000.0	0.0	laminar
0.0506501048346889	0.1147428056677419	363920.1806988166	0.1932577593969861	112.43016773261913	0.907211426284797	0.096196452331363	2.4361972779606	0.0020257475790659	267.79940496549074	1000000000.0	0.0	laminar
0.0739026679245839	0.1632868568057734	517883.2963747615	0.1005408206484053	118.1126263413937	0.9530638116653009	0.1171339183061391	2.417720290527037	0.0016279667442497	341.4097509388254	1000000000.0	0.0	laminar
0.1011191421095574	0.2195305250490136	696266.645039893	0.0275011037343019	122.3711244779754	0.9874261029100072	0.1379820184358799	2.424469996959095	0.0013256100456726	415.5761314584663	1000000000.0	0.0	laminar
0.1320340524549544	0.2829946292313039	897550.4477803087	-0.0329494806471806	125.7658705528162	1.0148186834828234	0.1583164141844545	2.4315110016108274	0.0011162639250132	488.9845088286731	1000000000.0	0.0	laminar
0.1663	0.3531	11199	-	128.55	1.0373	0.1781	2.4380	0.0009	561.44	1000	0.0	la

455116316067	01330536502	02.021447278	0.084089705795448	355423837776	1281066783	594942843166	065902437167	639488700787	44888694215	000000.0		minar
0.2037193715819191	0.4291984666370509	1361252.9572136996	-0.1271718327981587	130.84602165898744	1.0558109831806453	0.197699228275379	2.445991443303034	0.0008457807666547	633.1623487040873	1000000000.0	0.0	laminar
0.2437940563058938	0.5105788670995084	1619360.3816338223	-0.1623629394921668	132.6826339364907	1.070630810255603	0.2172106415197521	2.457527586349515	0.0007486793655584	704.5383141184823	1000000000.0	0.0	laminar
0.2861858395053481	0.5964965491145282	1891858.323288388	-0.1893005509377956	134.06756513307926	1.081805972860994	0.2370306584331561	2.4743496124265905	0.0006646981233122	776.1142268619838	1000000000.0	0.0	laminar
0.3304942587938113	0.6861799377514877	2176299.6423629136	-0.2074614129264459	134.99133760689108	1.089259994859098	0.2575364781741493	2.498514016582062	0.000588804342985	848.5256312975545	1000000000.0	0.0	laminar
0.3763073240029427	0.7788417676108137	2470187.43489202	-0.2164234882783799	135.444319635477	1.092915156818939	0.2791339566329301	2.532538778118937	0.0005175865431233	922.4806156020884	1000000000.0	0.0	laminar
0.4232061945767489	0.8736854550834547	2770995.2431229902	-0.2160467187841336	135.42531406030483	1.092761798736255	0.3022483735904362	2.5797255541677053	0.0004485918812306	998.74206630817	1000000000.0	0.0	laminar
0.4707690743351984	0.9699081665724224	3076176.7867379705	-0.2065757407003436	134.94646911402626	1.0888979460404575	0.3273207632769284	2.646114850380423	0.0003797129322574	1078.1244454137102	1000000000.0	0.0	laminar
0.5185741934072935	1.0667013259415112	3383167.5722867246	-0.1886639039134669	134.035037860133	1.081543506707311	0.3548169854934218	2.7400551349838134	0.0003090108129409	1161.5284896385317	1000000000.0	0.0	laminar
0.5662018980247188	1.1632497261198471	3689382.075536924	-0.1633246247731674	132.73238458462868	1.0710322537237225	0.3852576933701168	2.882958974786737	0.0002333698739791	1250.037146776807	1000000000.0	0.0	laminar
0.6132360223540745	1.2587306107483358	3992210.828808826	-0.1318263760435061	131.09076001942188	1.0577858040095562	0.5238539560392241	1.4616082744355032	0.0034336779370803	1677.4363169770668		1.0	turbulent
0.659264843410772	1.352314020522654	4289021.519442441	-0.0955554440900539	129.1685290409386	1.0422751101912753	0.7120474260773553	1.437755866041888	0.0032943583739482	2249.63028472964		1.0	turbulent
0.703881995	1.443165382	4577167.201	-0.0558	127.0245984	1.024975497	0.903096814	1.431907233	0.003132193	2810.003826		1.0	turbu

0989186	4649215	974402	722642565504	3724	640108	0600156	1027094	2611275	077468			lent
0.7466877251231454	1.5304508477617995	4854003.227713577	-0.0139822545756955	124.71270106297396	1.0063205426878044	1.097853101058361	1.4319321683903836	0.0029857866674626	3359.058916578512		1.0	turbulent
0.7872908170545605	1.613345372850863	5116912.874848304	0.0291671591716043	122.27595559534572	0.9866581746974142	1.2977491111548691	1.4354633744431773	0.0028530143509883	3899.385795167505		1.0	turbulent
0.8253113815520968	1.6910430987466305	5363340.267688533	0.0729620688551411	119.74177812267358	0.9662096171100126	1.5043067503768206	1.4418959358019043	0.0027289613393173	4433.658480593429		1.0	turbulent
0.8603845694278387	1.7627692889369593	5590828.239095532	0.1171783658209099	117.116790903076	0.9450283056568984	1.719722409754261	1.4513497418842145	0.0026085327664561	4965.739145610218		1.0	turbulent
0.892165104533994	1.827792963070222	5797058.399693135	0.1620354808785207	114.38078351124776	0.9229511601867496	1.9479120890836832	1.4645771349925705	0.0024858746262897	5502.6179262836085		1.0	turbulent
0.9203324048391728	1.8854393957576685	5979890.779276088	0.2082750082940114	111.47774743784215	0.8995262418590114	2.196314608687572	1.4832651879792376	0.0023530204422159	6057.550595211709		1.0	turbulent
0.9445959764010288	1.9351017924661469	6137400.858264567	0.2573139100148505	108.29966444597088	0.8738819396040883	2.479388530776198	1.510943442075729	0.0021973371289775	6655.862253399164		1.0	turbulent
0.9647007361335488	1.9762516523275704	6267912.434562116	0.3115835067696456	104.6522734793476	0.8444507395286962	2.8265811325281867	1.5556749396463485	0.0019955301286636	7347.739728569818		1.0	turbulent
0.980431941297911	2.008447529886631	6370025.412459643	0.3753690915848507	100.17042017489948	0.8082861707948193	3.304661215617957	1.6273605477399726	0.0017301072312149	8243.054819788396		1.0	turbulent

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated Tu)

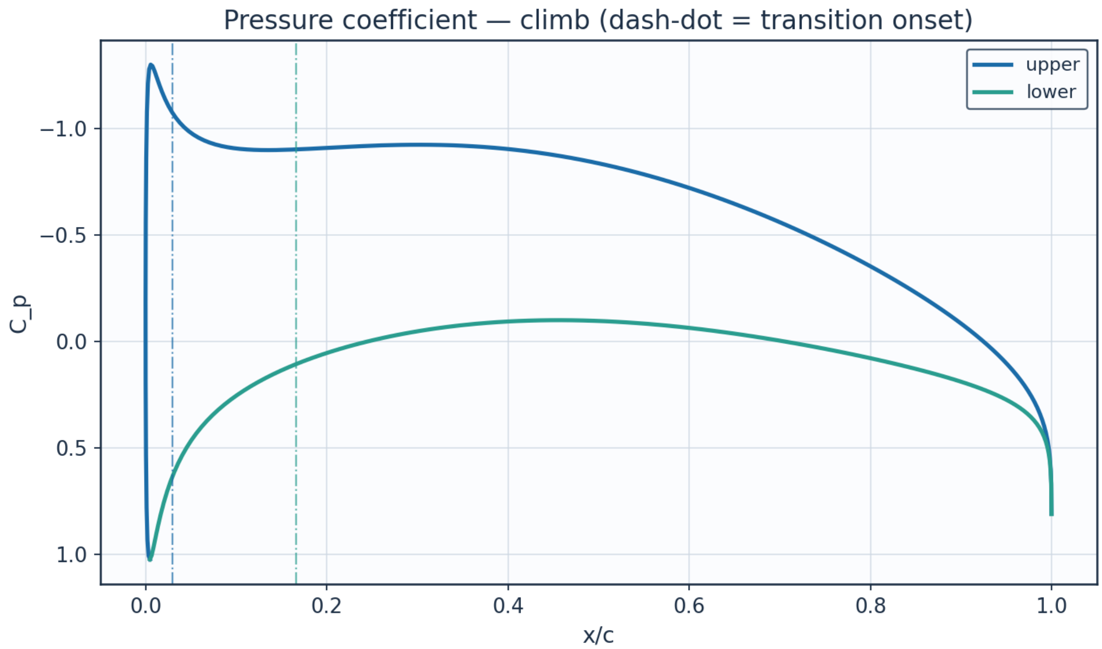


Fig. 18. Pressure coefficient C_p — climb.

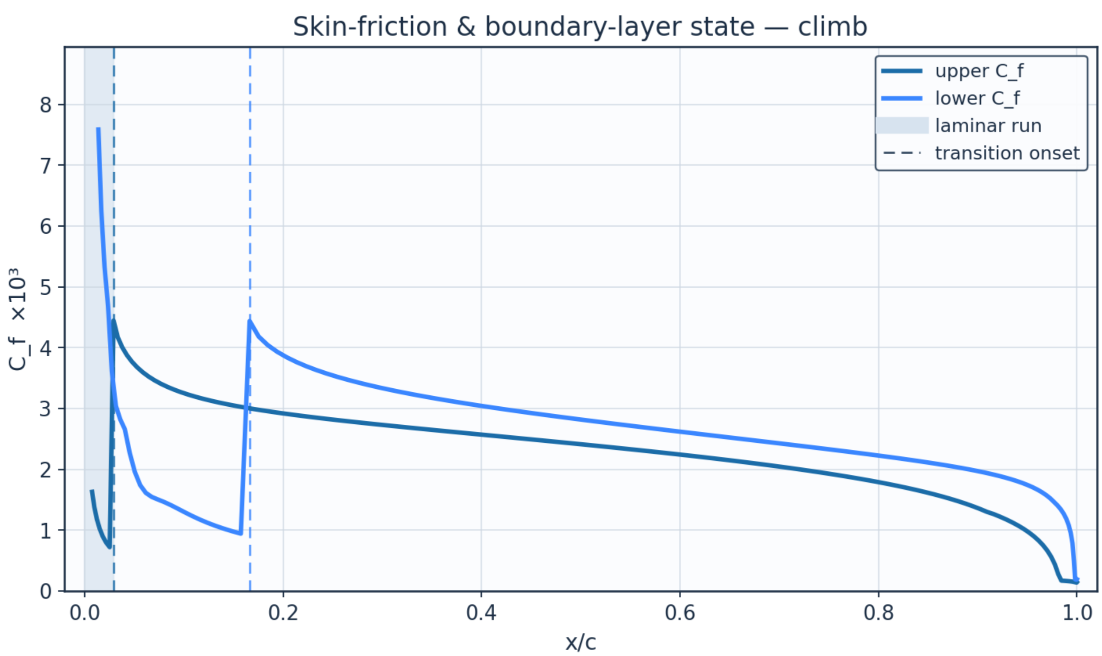


Fig. 19. Skin-friction C_f and laminar run — climb.

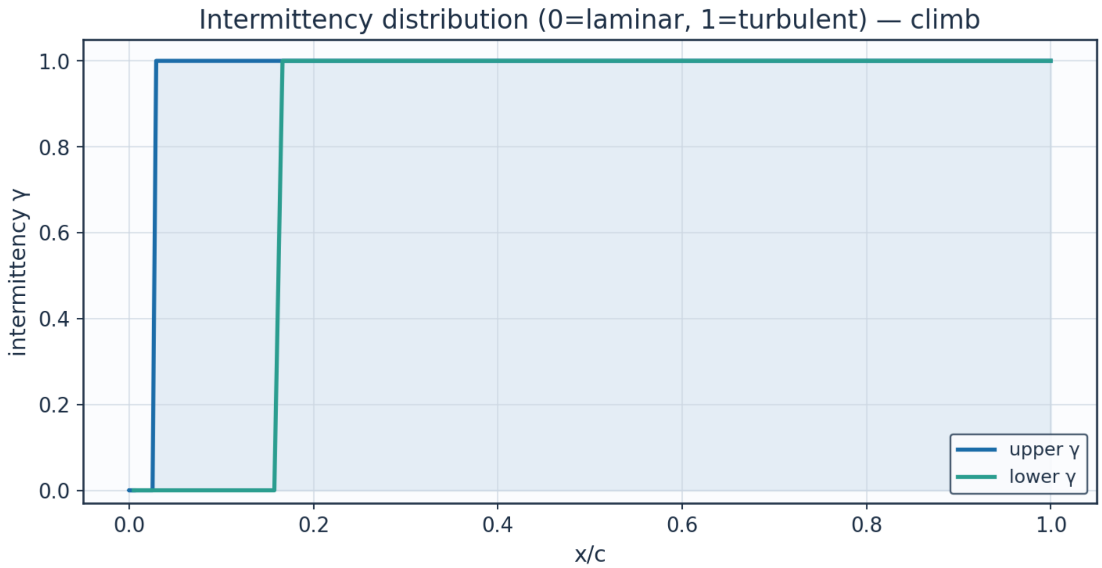


Fig. 20. Intermittency γ — climb.

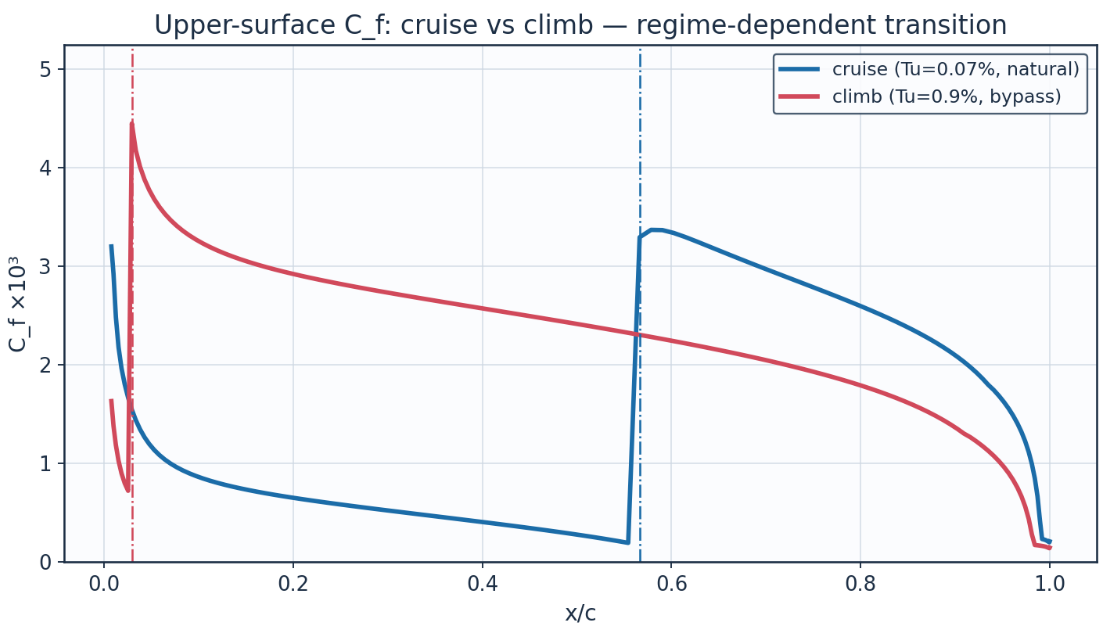


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1114	0.00568	-19.6	0.673	0.149
-2.0	0.0287	0.00456	6.3	0.65	0.4
-1.0	0.1683	0.00435	38.7	0.626	0.471
0.0	0.3077	0.00425	72.5	0.602	0.531
1.0	0.4472	0.00434	103.1	0.566	0.566
2.0	0.5872	0.00447	131.3	0.53	0.602
3.0	0.7281	0.00523	139.1	0.408	0.625

4.0	0.8704	0.00768	113.3	0.048	0.659
5.0	1.0146	0.00836	121.3	0.025	0.682
6.0	1.1612	0.00901	128.8	0.018	0.715
7.0	1.3111	0.00967	135.6	0.022	0.736
8.0	1.4654	0.01061	138.2	0.015	0.767

Table 17. Aerodynamic polar (aero_polar.csv).

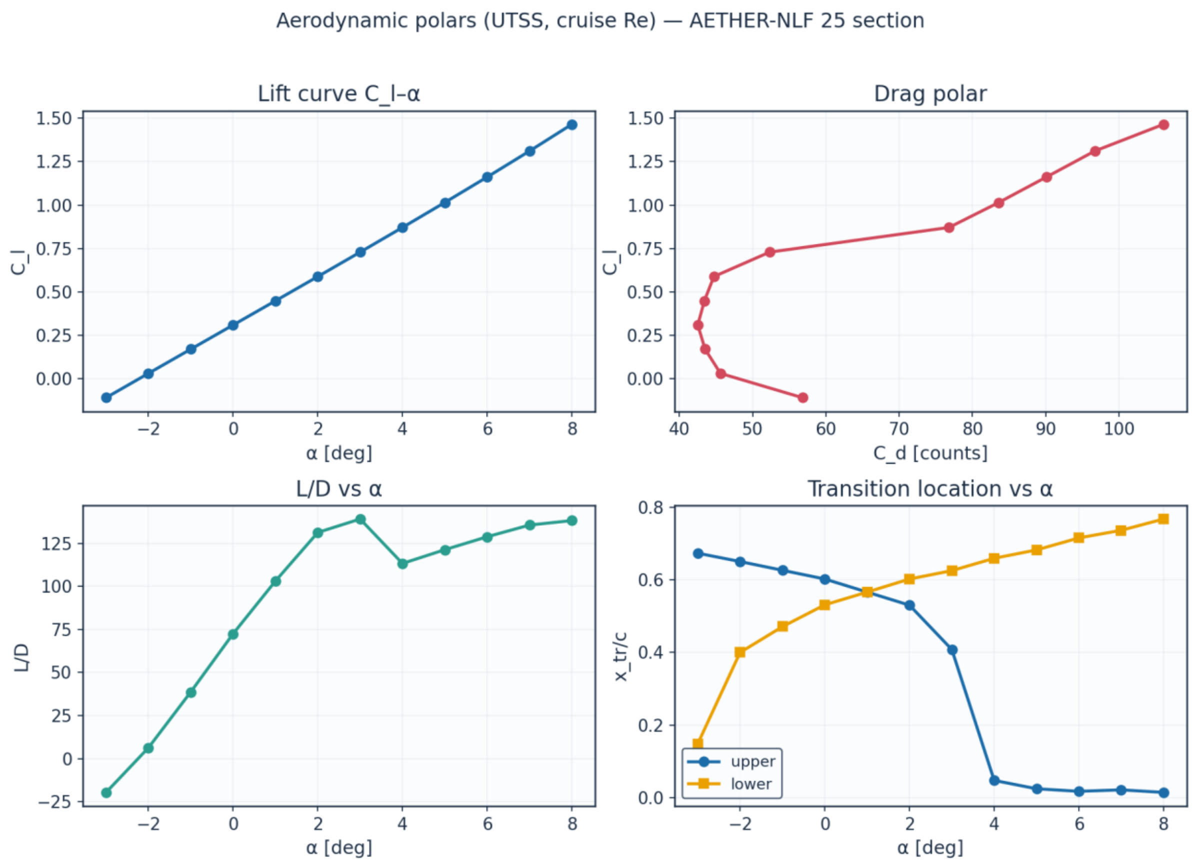


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.0	2.6	8246200.0	1.5	0.53	0.566	0.00431	0.548
0.089	0.757	2.484	7878900.0	1.23	0.542	0.554	0.00431	0.548
0.178	1.515	2.368	7511500.0	0.97	0.554	0.554	0.00428	0.554
0.267	2.272	2.253	7144200.0	0.7	0.566	0.542	0.0043	0.554
0.356	3.029	2.137	6776900.0	0.43	0.578	0.542	0.00428	0.56
0.445	3.786	2.021	6409500.0	0.16	0.59	0.531	0.00431	0.56
0.535	4.544	1.905	6042200.0	-0.1	0.602	0.531	0.00431	0.566
0.624	5.301	1.789	5674900.0	-0.37	0.614	0.519	0.00435	0.566
0.713	6.058	1.673	5307600.0	-0.64	0.626	0.519	0.00436	0.572
0.802	6.815	1.558	4940200.0	-0.91	0.684	0.507	0.0042	0.595
0.891	7.573	1.442	4572900.0	-1.17	0.695	0.495	0.00428	0.595
0.98	8.33	1.326	4205600.0	-1.44	0.707	0.495	0.00433	0.601

Table 18. Span-wise distribution (spanwise_distribution.csv).

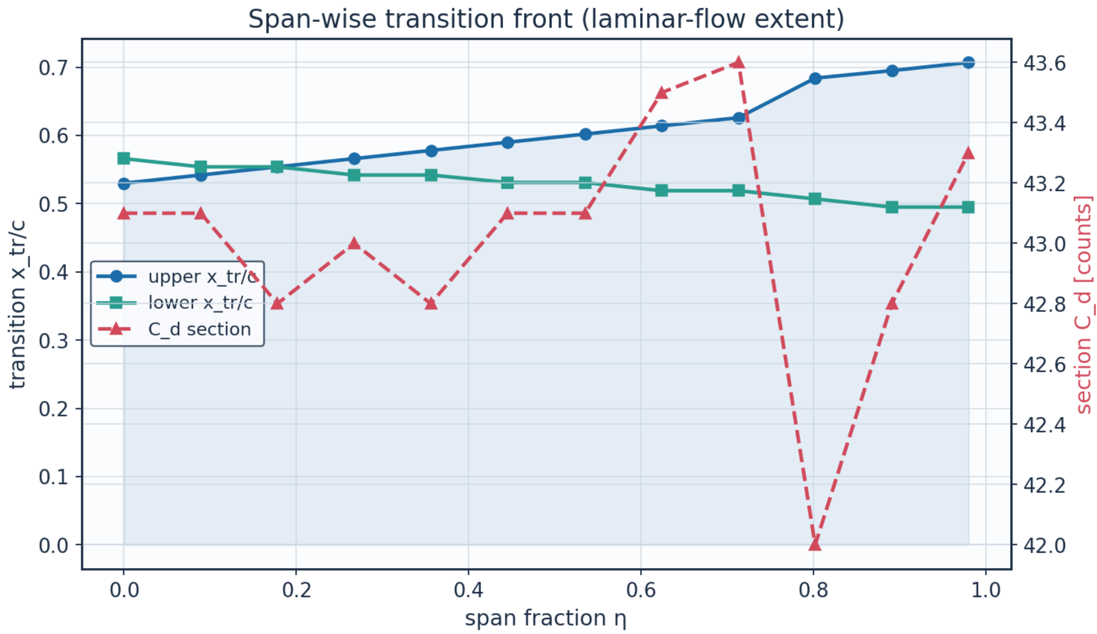


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

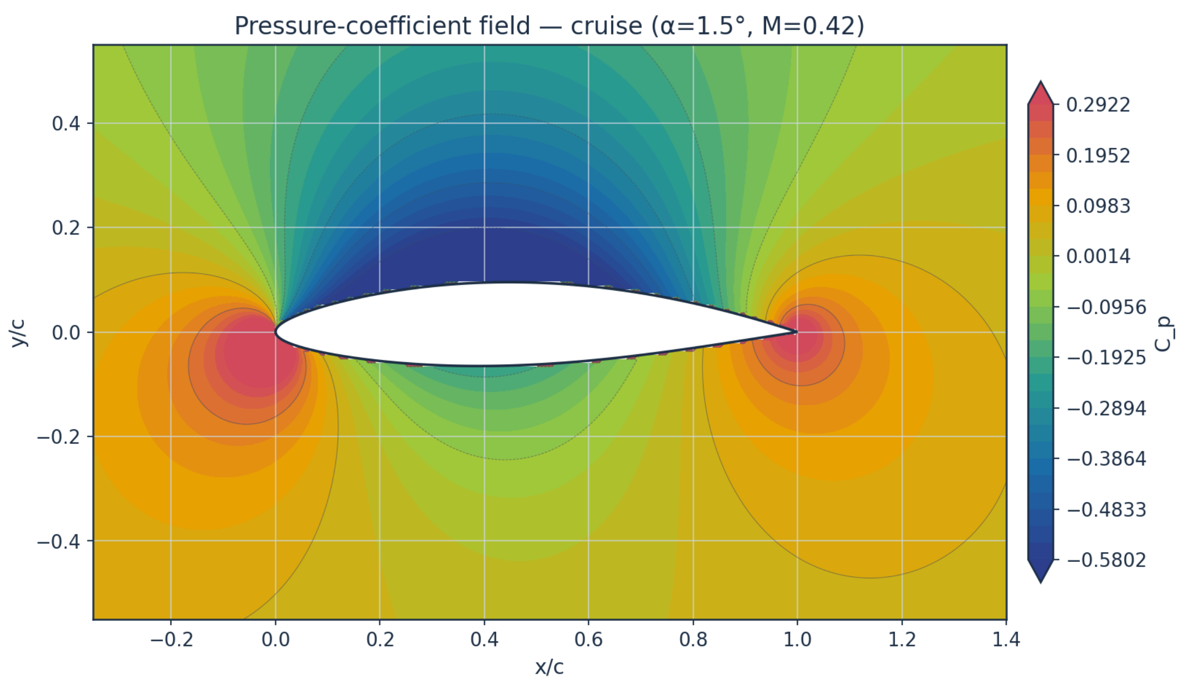


Fig. 24. Pressure-coefficient contour — cruise.

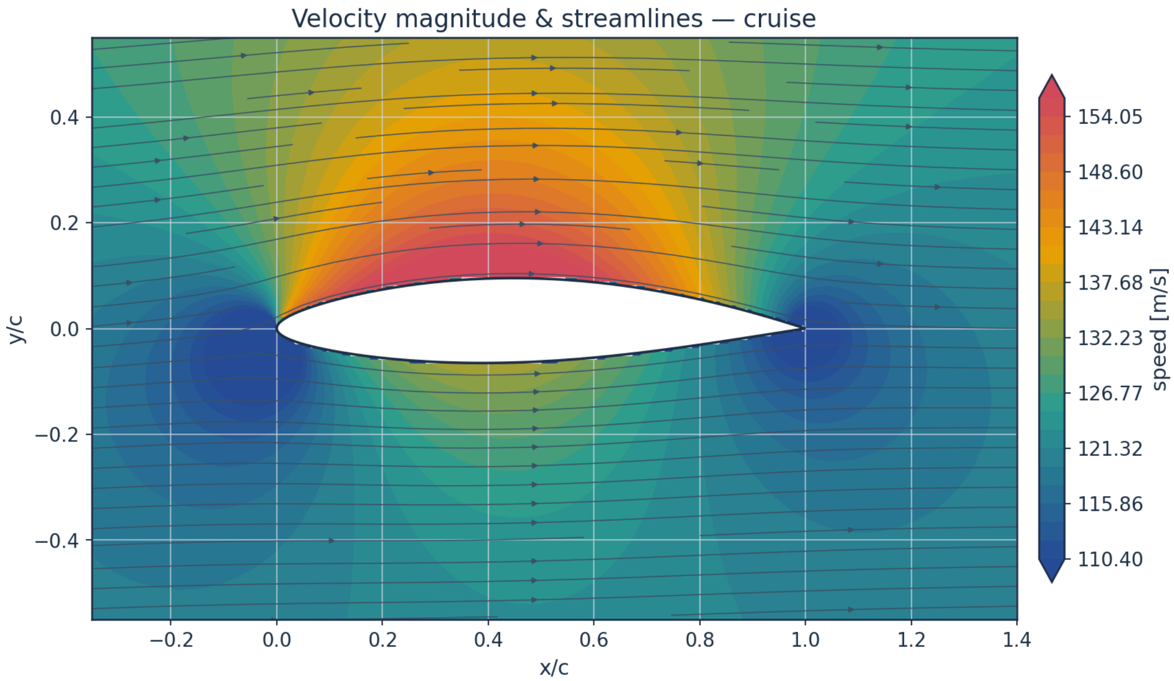


Fig. 25. Velocity magnitude and streamlines — cruise.

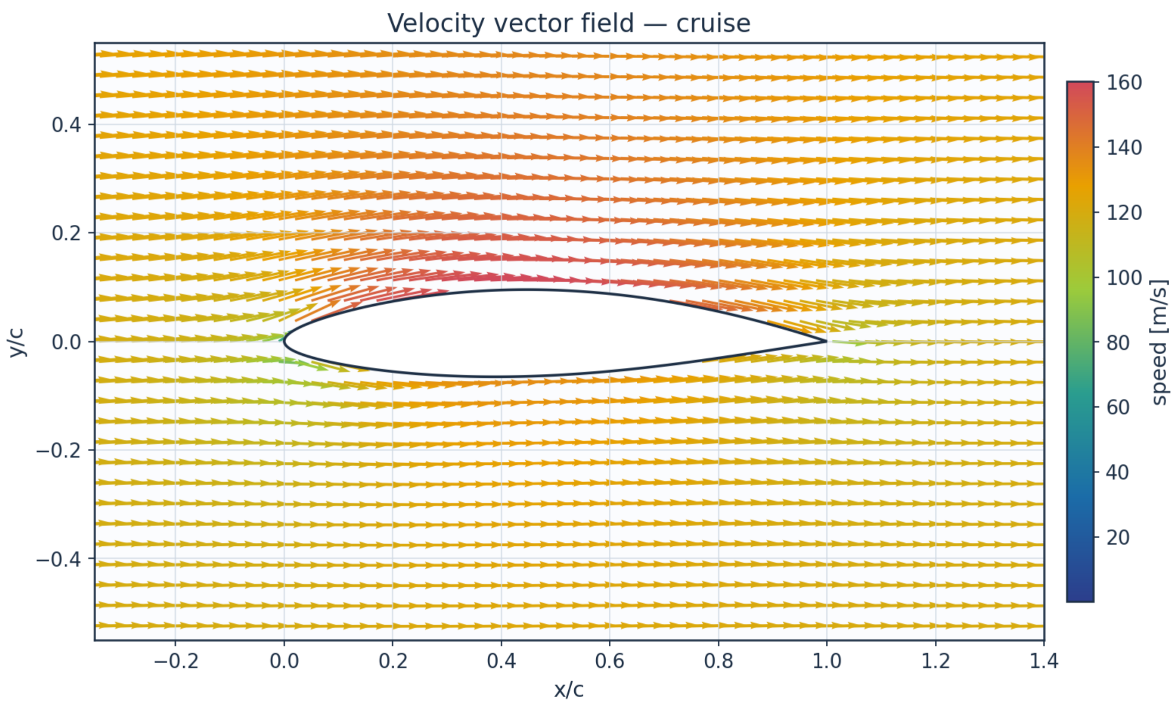


Fig. 26. Velocity vector field — cruise.

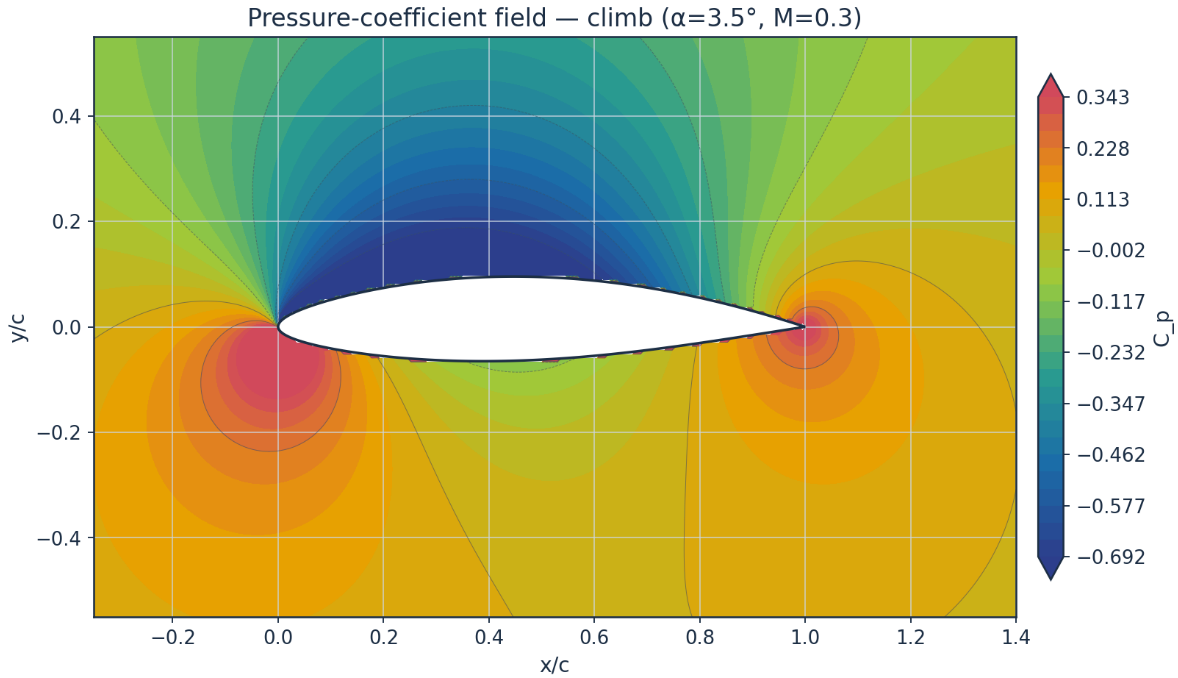


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

Temperature profiles are obtained from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

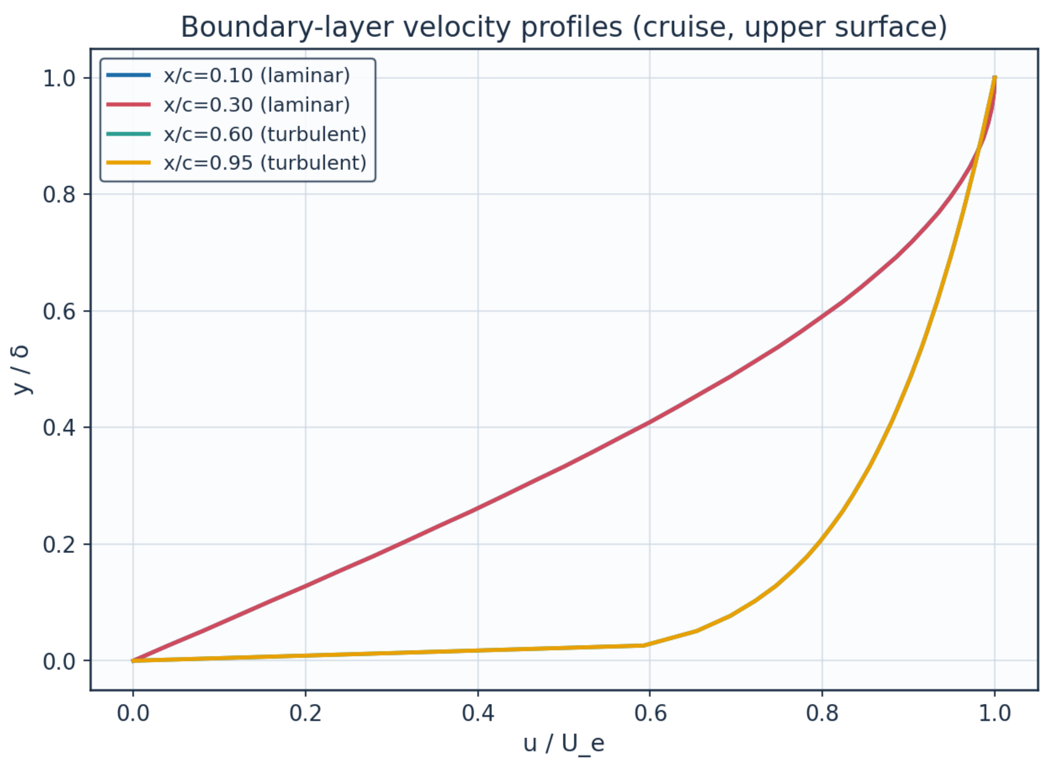


Fig. 28. Boundary-layer velocity profiles.

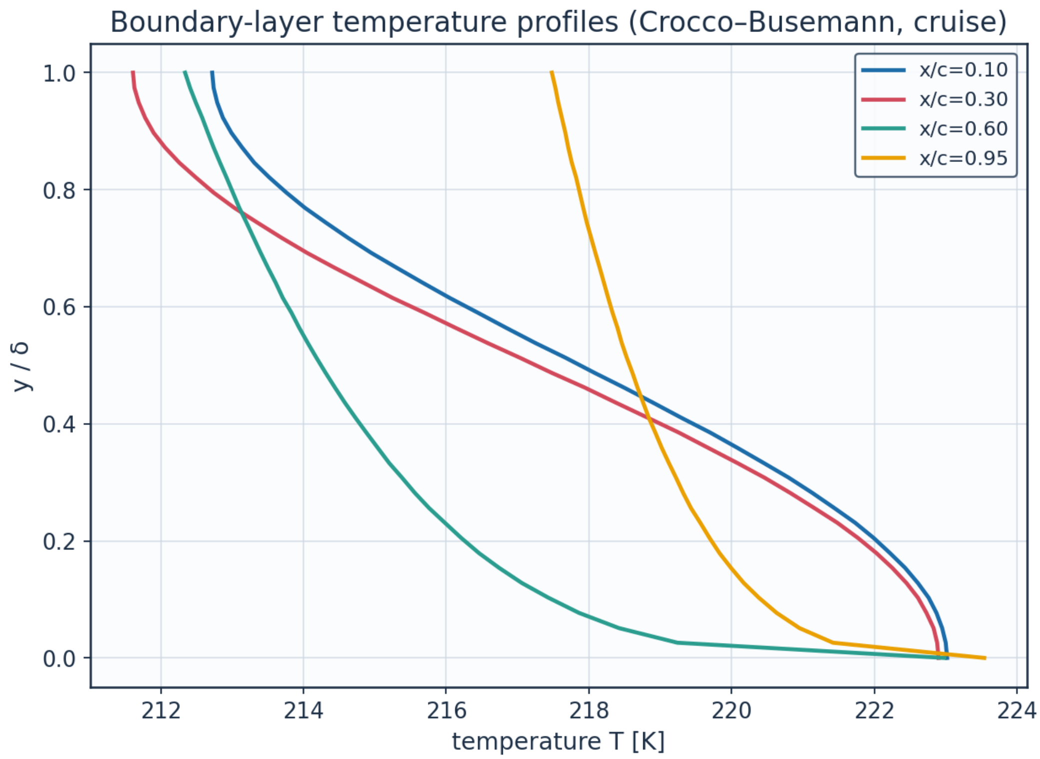


Fig. 29. Boundary-layer temperature profiles (K).

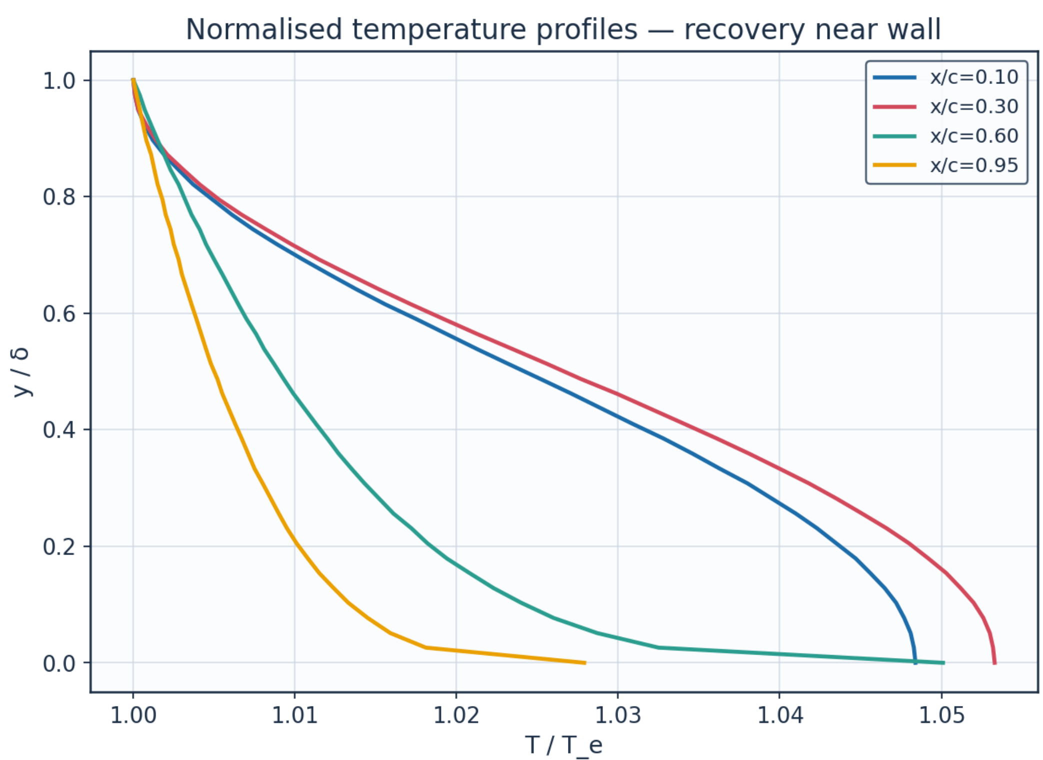


Fig. 30. Normalised temperature profiles T/T_e.

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K	Me_edge	state
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02	0.521	laminar
x/c=0.10	0.1	0.128	0.2471	0.2	1.0465	222.61	0.521	laminar
x/c=0.10	0.1	0.256	0.4942	0.392	1.041	221.44	0.521	laminar
x/c=0.10	0.1	0.385	0.7413	0.5681	1.0328	219.7	0.521	laminar
x/c=0.10	0.1	0.513	0.9884	0.7212	1.0232	217.67	0.521	laminar
x/c=0.10	0.1	0.641	1.2355	0.8452	1.0138	215.67	0.521	laminar
x/c=0.10	0.1	0.769	1.4826	0.935	1.0061	214.02	0.521	laminar
x/c=0.10	0.1	0.897	1.7297	0.9871	1.0012	212.99	0.521	laminar
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9	0.547	laminar
x/c=0.30	0.3	0.128	0.4091	0.2	1.0512	222.45	0.547	laminar
x/c=0.30	0.3	0.256	0.8182	0.392	1.0451	221.16	0.547	laminar
x/c=0.30	0.3	0.385	1.2272	0.5681	1.0361	219.26	0.547	laminar
x/c=0.30	0.3	0.513	1.6363	0.7212	1.0256	217.03	0.547	laminar
x/c=0.30	0.3	0.641	2.0454	0.8452	1.0152	214.84	0.547	laminar
x/c=0.30	0.3	0.769	2.4545	0.935	1.0067	213.03	0.547	laminar
x/c=0.30	0.3	0.897	2.8635	0.9871	1.0014	211.9	0.547	laminar
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98	0.531	turbulent
x/c=0.60	0.6	0.128	0.7901	0.7457	1.0223	217.06	0.531	turbulent
x/c=0.60	0.6	0.256	1.5803	0.8233	1.0161	215.76	0.531	turbulent
x/c=0.60	0.6	0.385	2.3704	0.8724	1.012	214.88	0.531	turbulent
x/c=0.60	0.6	0.513	3.1605	0.909	1.0087	214.18	0.531	turbulent
x/c=0.60	0.6	0.641	3.9507	0.9384	1.006	213.61	0.531	turbulent
x/c=0.60	0.6	0.769	4.7408	0.9632	1.0036	213.1	0.531	turbulent
x/c=0.60	0.6	0.897	5.531	0.9847	1.0015	212.66	0.531	turbulent
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54	0.396	turbulent
x/c=0.95	0.95	0.128	4.8772	0.7457	1.0124	220.17	0.396	turbulent
x/c=0.95	0.95	0.256	9.7545	0.8233	1.009	219.43	0.396	turbulent
x/c=0.95	0.95	0.385	14.6317	0.8724	1.0067	218.93	0.396	turbulent

Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled).

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

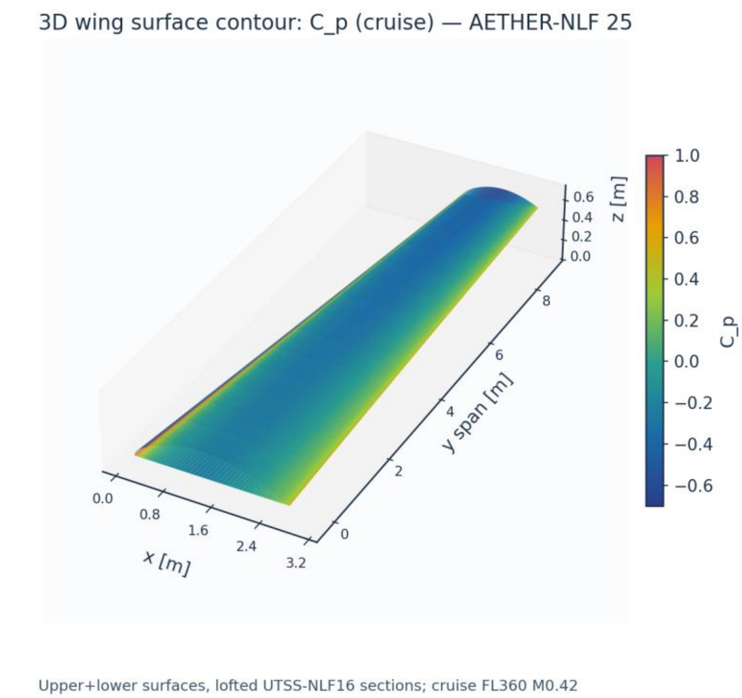


Fig. 31. 3-D wing surface contour — pressure C_p.

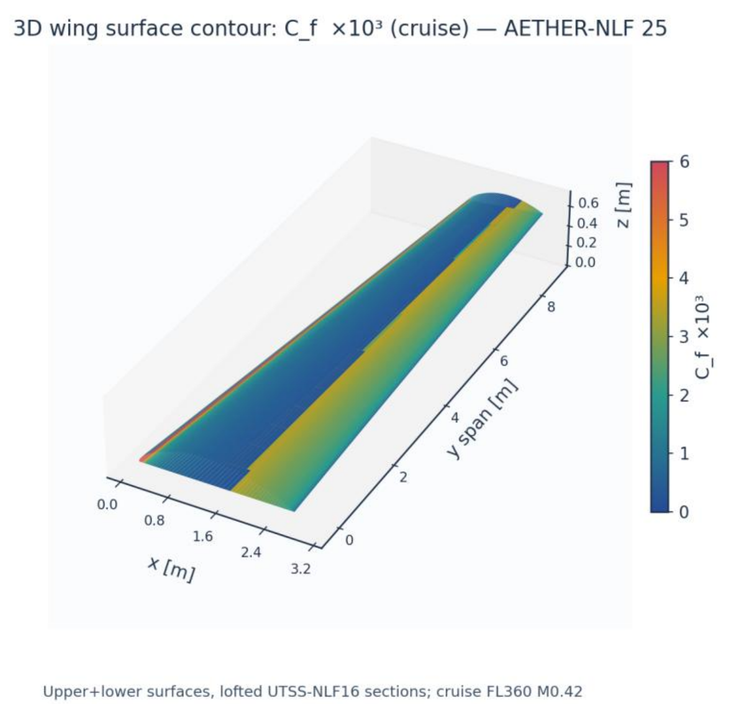


Fig. 32. 3-D wing surface contour — skin friction C_f (×10³).

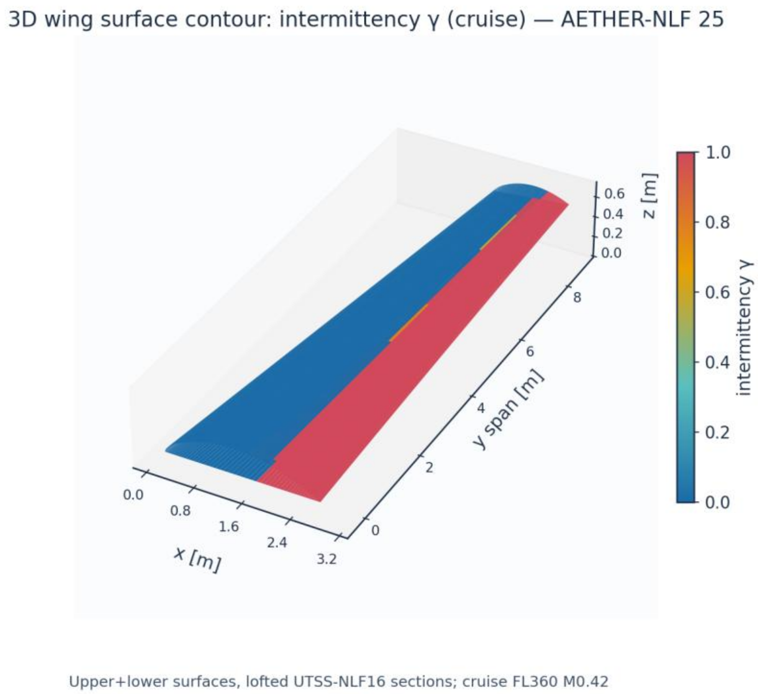


Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

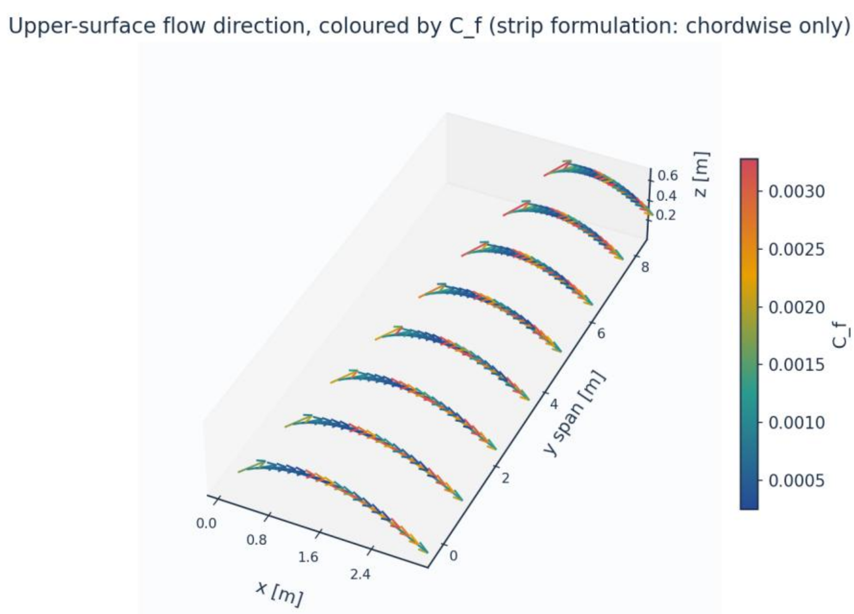


Fig. 34. Upper-surface flow direction coloured by C_f (chordwise only — the strip formulation carries no span-wise wall shear).

11. Validation and Calibration

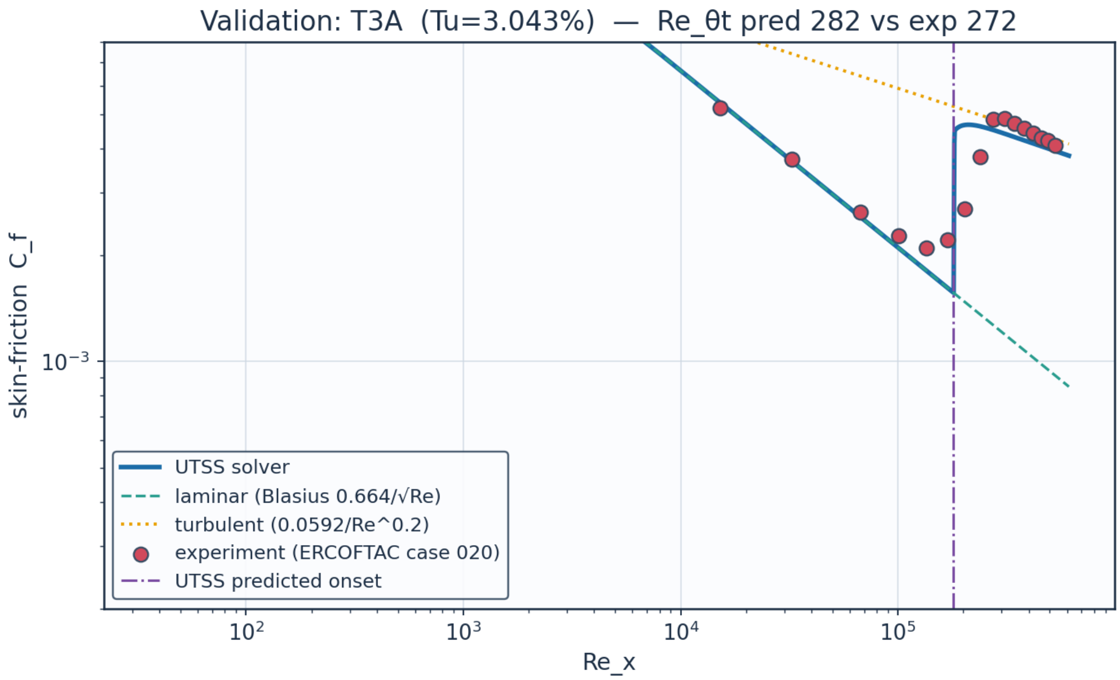
To qualify as universal, the solver is validated against every published dataset the kernel's four branches reach — five flat plates spanning 0.03 to 6 per cent free-stream turbulence (bypass, natural and separation-induced transition), 86 aerofoil conditions on the NLF(1)-0416 section, and two swept wings from different facilities and eras — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number Re_θt is the metric on the plates and the transition location x_tr/c on the aerofoil and the swept wings. The skin-friction error is reported separately over the laminar run and over the turbulent run, on the stations where the measurement and the prediction are in the same state; pooled across transition it measures the onset error a second time, in the wrong units, because a plate whose onset is early by 16 % is then charged with the whole laminar-to-turbulent step in C_f over the interval between the two onsets.

case	Tu_inlet_pct	L_turb_mm	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp	Re_x_tr_pred	Cf_err_laminar_pct	n_pts_laminar	Cf_err_turbulent_pct	n_pts_turbulent	Cf_err_all_pts_pct	Re_theta_meas_over_blasius	mechanism
ERCOFTAC T3A flat plate (bypass transition)	3.043	1.5299999999999998	272.3	282.1	3.6	134800.0	180400.0	3.8	4	8.2	9	12.7	1.117	bypass
ERCOFTAC T3A- flat plate (low-Tu bypass)	0.874	0.98	818.8	683.6	-16.5	1443000.0	1060000.0	4.2	8		0	216.9	1.027	bypass
ERCOFTAC T3	5.952	3.83	181.3	168.0	-7.3	59100.0	64050.0	8.2	3	9.7	11	10.1	1.123	bypass

B flat plate (high-Tu bypass)														
ERCOFTAC T3C4 flat plate (laminar separation bubble)	2.11		381.3	271.4	-28.8	178500.0	169200.0	29.7	10		0	220.9	1.359	separation
Schubauer & Skramstad flat plate (natural transition)	0.03		1100.0	1161.6	5.6	2800000.0	3060000.0		0		0			TS-natural

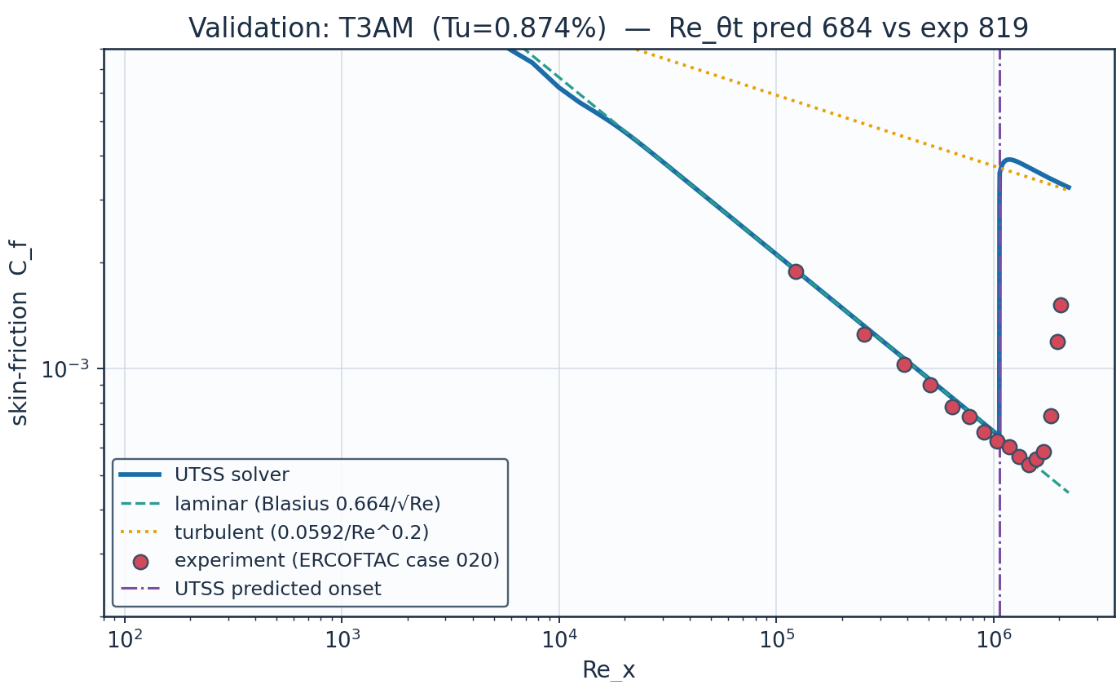
Table 20. Validation summary — predicted vs published $Re_{\vartheta t}$.

11.1 Flat plates



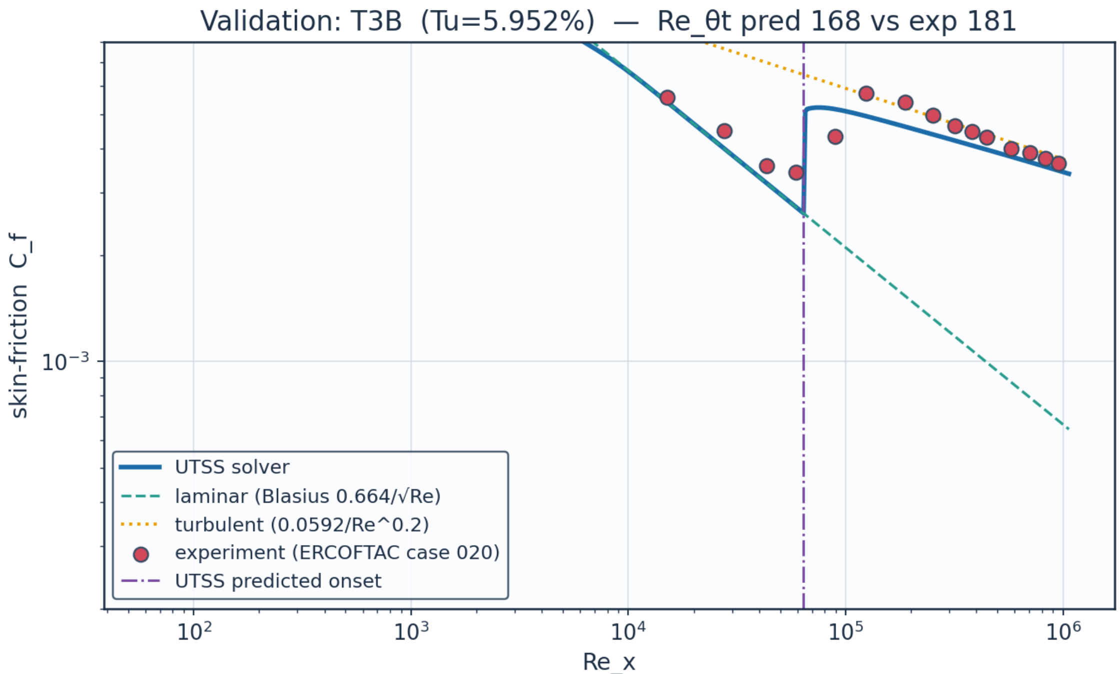
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 35. Validation — ERCOFTAC T3A flat plate (Tu = 3.0 %, bypass).



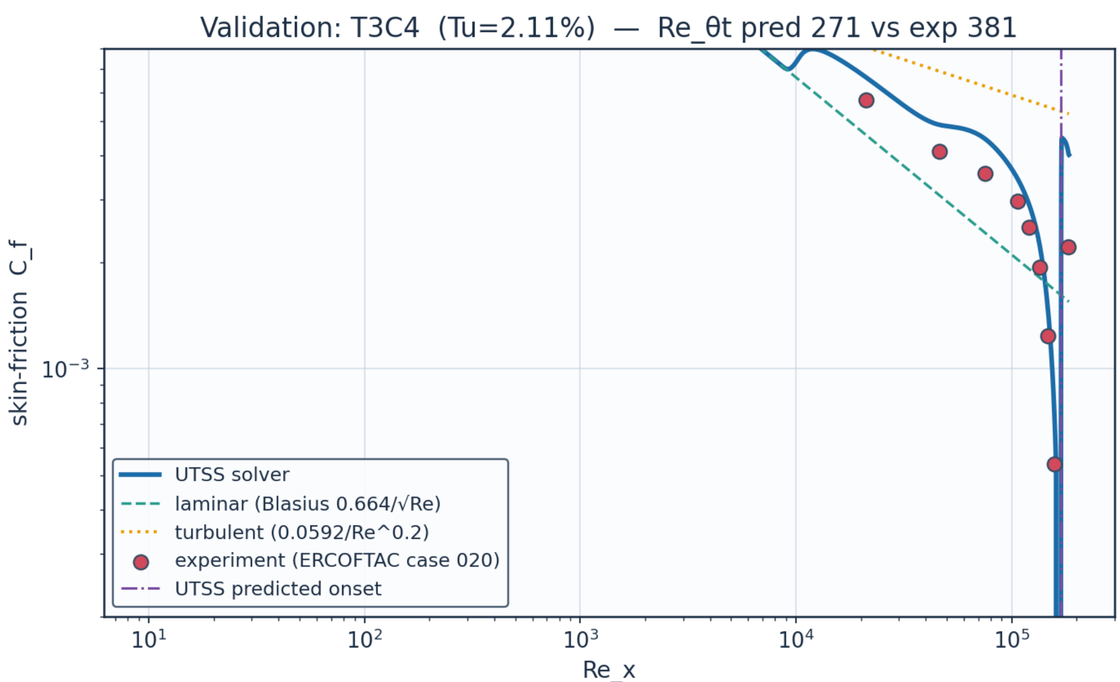
Source: Roach & Brierley (1990), ERCOFTAC T3A-; ERCOFTAC Classic Collection Case 020....

Fig. 36. Validation — ERCOFTAC T3A⁻ flat plate (Tu = 0.87 %, bypass).



Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 37. Validation — ERCOFTAC T3B flat plate (Tu = 6.0 %, bypass).



Source: Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-grad...

Fig. 38. Validation — ERCOFTAC T3C4 flat plate (laminar separation bubble).

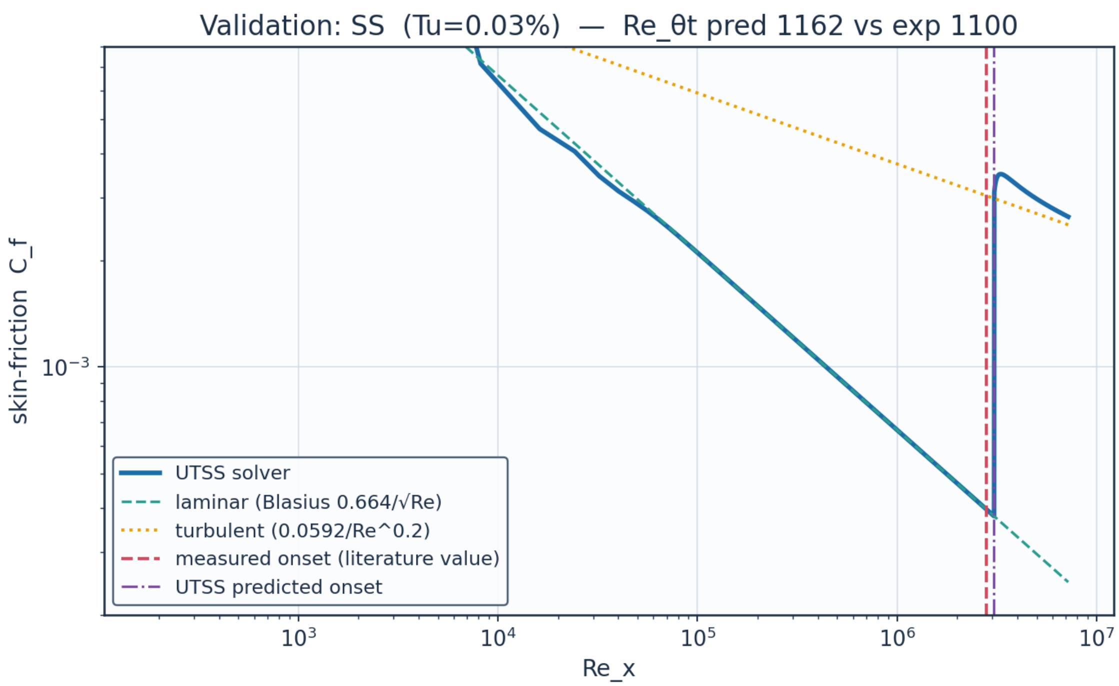


Fig. 39. Validation — Schubauer & Skramstad natural transition (Tu = 0.03 %).

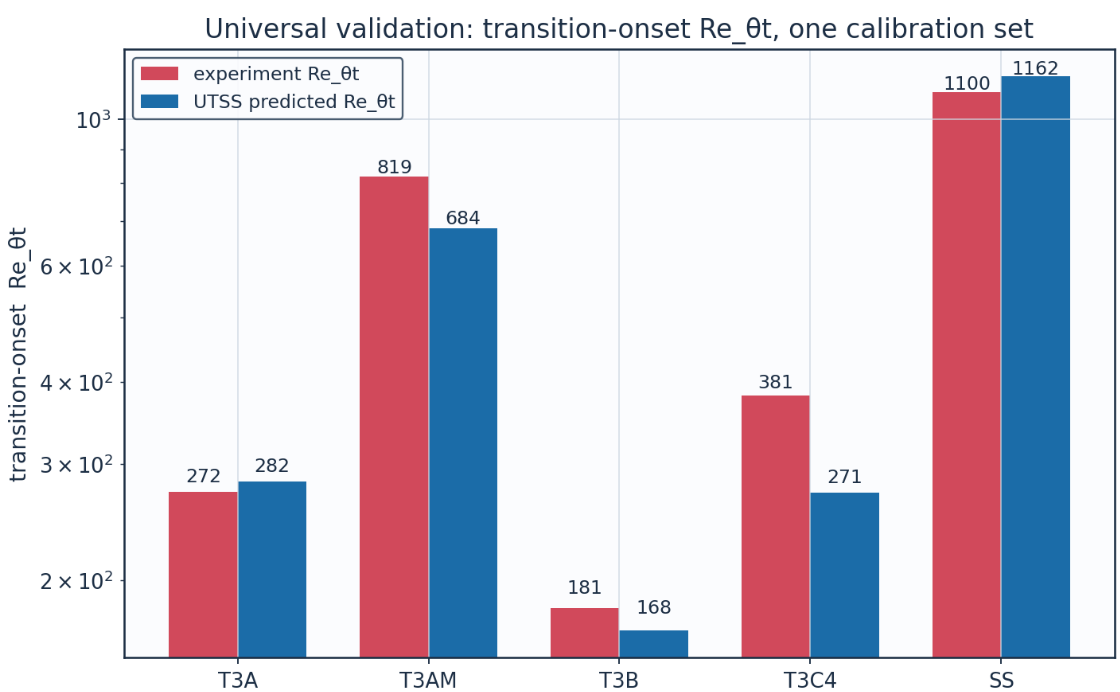


Fig. 40. Universal validation — Re_ϑt across all five plates.

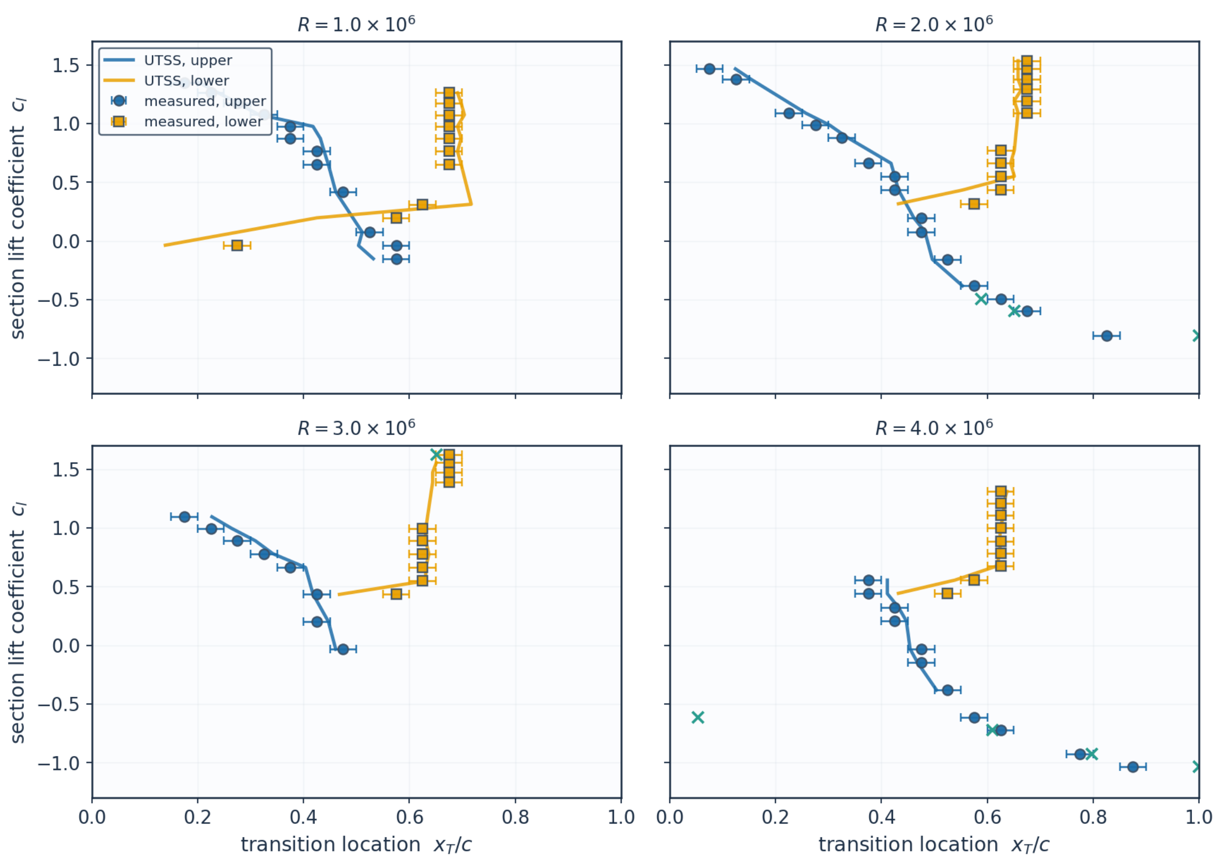
11.2 NLF(1)-0416 aerofoil — 86 transition locations

The largest single body of evidence in this work, and the one on which nothing is calibrated. Transition locations were digitised from Fig. 9 of the source report at four chord Reynolds numbers on both surfaces, and each condition is matched by trimming the incidence to the measured lift coefficient. The experiment brackets transition between adjacent orifices 0.05c apart, so its own uncertainty is ±0.025c and a prediction inside that band cannot be distinguished from the measurement. Both the natural (TS) and the separation-induced branches are selected on this set.

set	points	mean_abs_err_c	bias_c	within_bracket	within_bracket_pct
Upper surface	46	0.0409	0.0025	25	54.3
Lower surface	40	0.0332	-0.0148	27	67.5
All	86	0.0373	-0.0055	52	60.5
Conditions the method accepts	83	0.0288	-0.0031	52	62.7

Table 21. NLF(1)-0416 error statistics by surface (aerofoil_nlf0416_summary.csv).

NLF(1)-0416 transition location, M = 0.10 — 86 points digitised from NASA TP-1861, Fig. 9



Bars show the +/-0.025c orifice-pitch bracket within which the experiment localises transition. Nothing is calibrated on this set.

Fig. 41. Transition location against lift coefficient at four chord Reynolds numbers, measurement bars = the ±0.025c orifice bracket.

Re_c	surface	c_l_exp	alpha_deg	c_l_solver	x_tr_c_exp	x_tr_c_pred	err_c	err_pct	within_bracket	degenerate	burst	le_sep	x_sep_c	mechanism
1000000.0	upper	1.35	6.483	1.3503	0.175	0.203	0.028	16.1	False	False	False	False		TS-natural
1000000.0	upper	1.174	5.036	1.1743	0.275	0.276	0.001	0.4	True	False	False	False		TS-natural
1000000.0	upper	0.976	3.413	0.9762	0.375	0.418	0.043	11.5	False	False	False	False	0.335	separation
1000000.0	upper	0.765	1.687	0.765	0.425	0.439	0.014	3.4	True	False	False	False	0.356	separation
1000000.0	upper	0.419	-1.132	0.4187	0.475	0.461	-0.014	-3.0	True	False	False	False	0.383	separation
1000000.0	upper	-0.039	-4.846	-0.0387	0.575	0.504	-0.071	-12.4	False	False	False	False		TS-natural
1000000.0	lower	1.262	5.759	1.2623	0.675	0.691	0.016	2.4	True	False	False	False	0.582	separation
1000000.0	lower	1.075	4.224	1.0753	0.675	0.704	0.029	4.4	False	False	False	False	0.582	separation
1000000.0	lower	0.874	2.578	0.8741	0.675	0.698	0.023	3.4	True	False	False	False	0.575	separation
1000000.0	lower	0.651	0.757	0.6508	0.675	0.698	0.023	3.4	True	False	False	False	0.568	separation
1000000.0	lower	0.197	-2.935	0.1968	0.575	0.425	-0.15	-26.0	False	False	False	False		TS-natural
2000000.0	upper	1.466	7.438	1.4661	0.075	0.124	0.049	66.0	False	False	False	False		TS-natural
2000000.0	upper	1.089	4.339	1.0893	0.225	0.257	0.032	14.3	False	False	False	False		TS-natural
2000000.0	upper	0.883	2.652	0.8831	0.325	0.335	0.01	3.2	True	False	False	False		TS-natural
2000000.0	upper	0.551	-0.058	0.5507	0.425	0.425	0.0	0.1	True	False	False	False	0.369	separation
2000000.0	upper	0.198	-2.927	0.1978	0.475	0.461	-0.014	-3.0	True	False	False	False	0.404	separation
2000000.0	upper	-0.156	-5.791	-0.1552	0.525	0.496	-0.029	-5.4	False	False	False	False		TS-natural
2000000.0	upper	-0.491	-8.514	-0.491	0.625	0.589	-0.036	-5.8	False	False	False	False		TS-natural
2000000.0	upper	-0.807	-11.077	-0.8069	0.825	1.0	0.175	21.2	False	True	True	False	0.947	none(laminar)
2000000.0	lower	1.469	7.463	1.4691	0.675	0.658	-0.017	-2.5	True	False	False	False	0.589	separation
2000000.0	lower	1.294	6.022	1.2943	0.675	0.665	-0.01	-1.5	True	False	False	False	0.589	separation
2000000.0	lower	1.093	4.372	1.0933	0.675	0.658	-0.017	-2.5	True	False	False	False	0.582	separation
2000000.0	lower	0.663	0.855	0.6628	0.625	0.644	0.019	3.1	True	False	False	False	0.568	separation

2000000.0	lower	0.434	-1.01	0.4337	0.625	0.553	-0.072	-11.4	False	False	False	False		TS-natural
3000000.0	upper	1.094	4.38	1.0943	0.175	0.227	0.052	29.5	False	False	False	False		TS-natural
3000000.0	upper	0.89	2.709	0.8901	0.275	0.309	0.034	12.2	False	False	False	False		TS-natural
3000000.0	upper	0.662	0.847	0.6618	0.375	0.404	0.029	7.8	False	False	False	False	0.363	separation
3000000.0	upper	0.202	-2.894	0.2018	0.425	0.447	0.022	5.1	True	False	False	False	0.404	separation
3000000.0	lower	1.623	8.733	1.6227	0.675	0.651	-0.024	-3.5	True	False	False	False	0.596	separation
3000000.0	lower	1.472	7.487	1.4721	0.675	0.644	-0.031	-4.6	False	False	False	False	0.589	separation

Table 22. NLF(1)-0416 point-by-point comparison (aerofoil_nlf0416.csv, sampled).

(table sampled to 30 rows; full data in 06_validation/aerofoil_nlf0416.csv)

11.3 Swept wings — the cross-flow branch

The cross-flow coefficient is set on the first of these two experiments and nothing is calibrated on the second, which is a different facility, section and era. The branch reproduces the calibration set to 14.7 % at the frozen constant C1 = 150. On the independent set that same constant gives 51.7 %, and C1 = 200 gives 18.4 % — so the criterion has the right functional form on both wings but not one critical value that serves both. The independent set is therefore tabulated at BOTH ends of that band: reporting only C1 = 150 understates what the criterion does here, and reporting only C1 = 200 would be a per-case re-tune of the kind this work is claiming not to need. The spread between the two columns is the limitation, stated rather than averaged away, and it is a receptivity difference — stationary cross-flow vortices are seeded by leading-edge roughness, which neither report documents. Nothing else in the model differs between the two columns.

Re_c	x_tr_c_exp	x_tr_c_pred	err_pct	Re_theta_at_onset	mechanism
1920000.0	0.78	0.695	-10.9	667.0	crossflow
2190000.0	0.73	0.605	-17.1	662.5	crossflow
2370000.0	0.58	0.559	-3.7	662.7	crossflow
2730000.0	0.45	0.488	8.5	664.5	crossflow
3270000.0	0.33	0.418	26.6	670.7	crossflow
3730000.0	0.3	0.364	21.4	665.6	crossflow

Table 23. Cross-flow validation, 45° swept NLF(2)-0415 (Dagenhart & Saric — the calibration set).

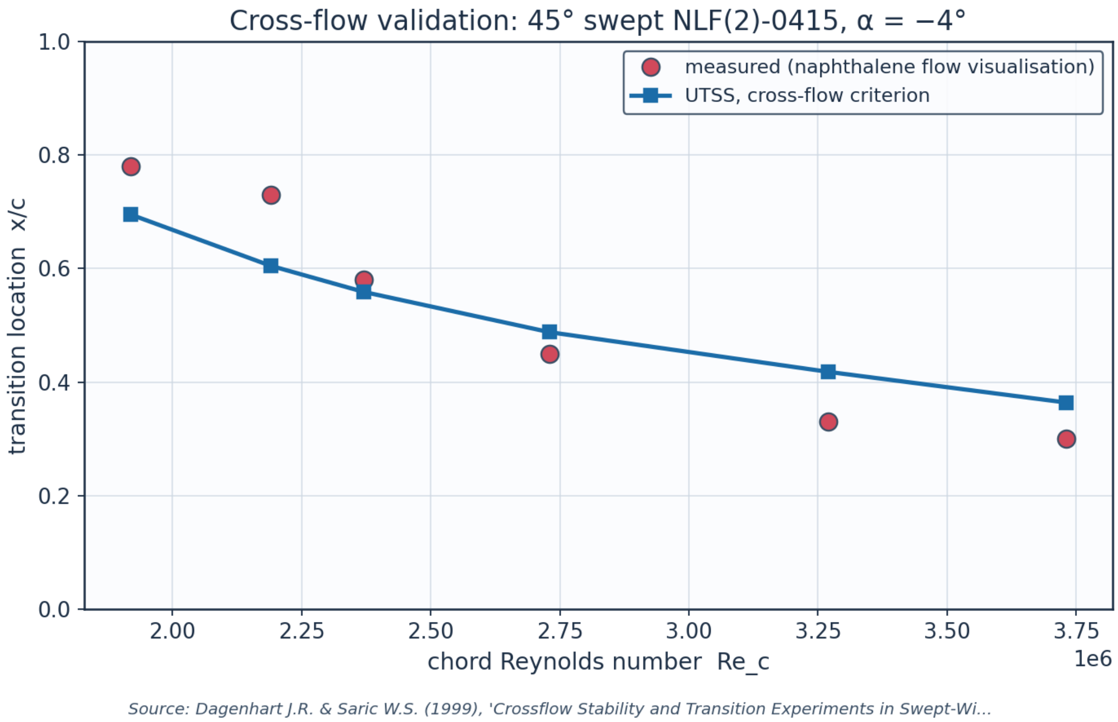


Fig. 42. Transition location against chord Reynolds number, 45° swept NLF(2)-0415.

sweep_deg	alpha_deg	Re_c	x_tr_c_exp	x_tr_c_pred_C1_150	err_pct_C1_150	mechanism	x_tr_c_pred_C1_200	err_pct_C1_200
20.0	0.0	27000000.0	0.207	0.093	-55.0	crossflow	0.161	-22.4
30.0	0.0	15000000.0	0.213	0.093	-56.2	crossflow	0.161	-24.6
40.0	0.0	12000000.0	0.2	0.098	-51.1	crossflow	0.161	-19.7
50.0	0.0	9500000.0	0.211	0.117	-44.4	crossflow	0.197	-6.8

Table 24. Independent swept-wing check, NACA 64(2)A015 (Boltz et al.) at both ends of the reported cross-flow band — nothing calibrated here.

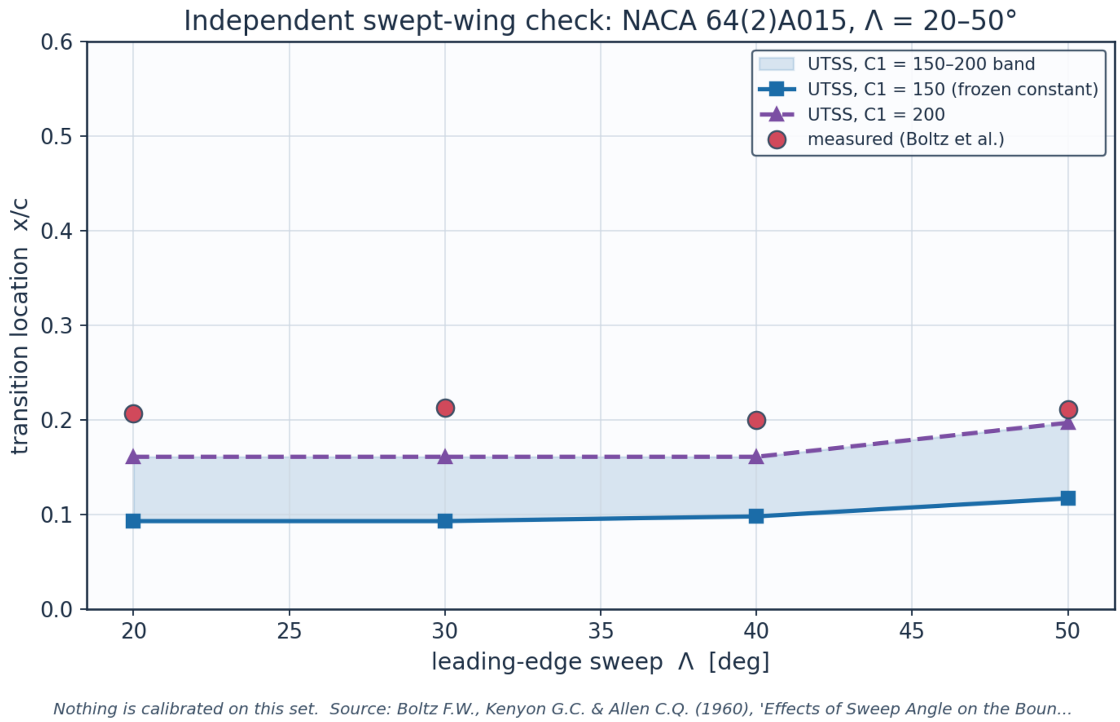


Fig. 43. Transition location against sweep angle, NACA 64(2)A015, with the C1 = 150–200 band shaded.

11.4 Ablations — what each element of the formulation is worth

Each of the three elements that distinguish this formulation is switched off in turn, everything else held fixed, and the same two datasets the natural branch reaches are re-run. The table is regenerated by gen_validation.py and is not quoted from memory.

configuration	SS_err_pct	mean_abs_err_c	within_bracket	points	within_bracket_pct
no bubble closure	5.6	0.0607	19	86	22.1
one-equation laminar march	-4.7	0.0388	44	86	51.2
Drela-Giles envelope	-6.6	0.0451	39	86	45.3
full model	5.6	0.0373	52	86	60.5

Table 25. Ablation study (ablations.csv): Schubauer-Skramstad onset error and the 86 aerofoil conditions, one closure removed at a time.

Calibration. The solver is calibrated through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, clamped to the 0.0008-0.0298 range over which that correlation is quoted, and so returns 8.18 for any stream quieter than 0.08 %. The SAME constants reproduce every validation dataset — five flat plates, two swept wings and 86 aerofoil conditions — demonstrating that no case-specific physics tuning is required, which is the essence of the universality claim.

11.5 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B (Tu=6%); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwieg H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Validation	ERCOFTAC T3A- flat plate	Roach & Brierley (1990), ERCOFTAC T3A- (Tu=0.87%); ERCOFTAC Classic Collection Case 020.
Validation	ERCOFTAC T3C4 flat plate (separation bubble)	Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-gradient series.
Validation	NLF(1)-0416 aerofoil transition (86 conditions)	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9; tunnel turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), NASA TP-2328, Fig. 25.
Validation	Swept-wing cross-flow (calibration set)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.
Validation	Swept-wing cross-flow (independent set)	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g).
Calibration	Linear stability / amplification	Orr (1907); Sommerfeld (1908);

	database	Gaster M. (1962), JFM 14, 222; Mack L.M. (1977), AGARD CP-224; Drela M. & Giles M.B. (1987), AIAA J. 25(10), 1347.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 26. Validation, calibration and case-study sources.

dataset	source
Dagenhart & Saric 45 deg swept NLF(2)-0415 (cross-flow)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.

Table 27. Cross-flow calibration dataset provenance.

dataset	source
Boltz, Kenyon & Allen NACA 64(2)A015 untapered wing	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g); transition locations digitised by the present author as x/c = R_trans/R.

Table 28. Independent swept-wing dataset provenance.

dataset	source	n_points	Tu_pct	orifice_pitch
McGhee et al. NLF(1)-0416 aerofoil, Langley LTPT	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9 (transition) and Table I (coordinates); turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), 'Recent Modifications and Calibration of the Langley Low-Turbulence Pressure Tunnel', NASA TP-2328, Fig. 25.	86	0.03	0.05

Table 29. Aerofoil dataset provenance.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- An amplification database in place of an envelope correlation: 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, tabulated against shape factor, momentum-thickness Reynolds number and frequency, with the march carrying one amplification factor per physical frequency. The stability solver reproduces the Blasius neutral point at Re_theta = 200 against the accepted 200.5.
- A separation-bubble closure that predicts a length rather than a point: the shear layer is carried across the dead-air region by the momentum integral with no wall stress — a step with no fitted constant, which reproduces the growth the T3C4 hot films record — and reattachment placed where the disturbance has amplified by the same N_crit used elsewhere, so the length scales with the disturbance environment.
- A two-equation laminar march whose closures are computed from the Falkner-Skan family rather than fitted, giving the shape factor a history. On the 86 aerofoil conditions it raises the number of predictions inside the experimental bracket from 44 to 52, improving both surfaces at once.
- Regime coverage with one constant set: transition-onset Re_theta_t predicted to +3.6 % on T3A, -16.5 % on T3A-, -7.3 % on T3B, -28.8 % on T3C4, +5.6 % on Schubauer & Skramstad, spanning 0.03-6 % free-stream turbulence intensity.
- An aerofoil validation built for this work: 86 transition locations digitised from Fig. 9 of NASA TP-1861 for the NLF(1)-0416 section, both surfaces, four chord Reynolds numbers and lift coefficients from -1.03 to +1.62, with nothing calibrated on them. Mean error 0.037 chord, and 52 of the 86 predictions fall inside the +/-0.025c bracket within which the experiment itself localises transition.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: 58 % laminar flow and 50 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated with a single calibration set against five flat plates, two independent swept-wing experiments and 86 aerofoil conditions, spans

0.03-6 % free-stream turbulence intensity. At cruise the wing achieves 58 % laminar flow and a 50 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver switches to the bypass route and predicts early transition, demonstrating regime coverage across the flight envelope.

Two limitations bound that claim and are stated here rather than left to be discovered. The cross-flow critical constant does not transfer between facilities: the two independent swept-wing experiments support the functional form of the criterion, and the measured data collapse on it, but reproducing the second requires a critical value 50 % larger than the first. And the method is an attached-flow formulation with an incidence envelope of about +/-8.5 deg on a real section; beyond that the laminar march separates near the leading edge and the result should not be relied on.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 105 (CSV: 44, PNG: 59, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV

04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/ablations.csv	CSV
06_validation/aerofoil_nlf0416.csv	CSV
06_validation/aerofoil_nlf0416_source.csv	CSV
06_validation/aerofoil_nlf0416_summary.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV

06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG
06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_aerofoil_nlf0416.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wal normal reconstruction stack (no volume mesh is generated)	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H*, l and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6
Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00438	-
Section Cd (counts)	43.8	counts
Section L/D	118.0	-
Wing lift (3-D estimate, C_L = 0.90 C_l)	43114.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.